A

Project Report

on

"NOISE REDUCTION OF LANDING GEAR USING ENHANCED MULTI-ELECTRODE DIELECTRIC BARRIER DISCHARGE PLASMA ACTUATOR"

(Subject Code: 15AEP85)

Submitted to

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
Belagavi-590018

In partial fulfillment of the requirement for award of the degree of

BACHELOR OF ENGINEERING
In
AERONAUTICAL ENGINEERING
(MAY-2020)

Submitted by

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CERTIFICATE

This is to certify that the project work entitled **"NOISE REDUCTION OF LANDING GEAR USING ENHANCED MULTI-ELECTRODE DIELECTRIC BARRIER DISCHARGE PLASMA ACTUATOR"** is a Bonafide work carried out by,

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In partial fulfillment for the award of degree of Bachelor of Engineering in Aeronautical Engineering of the Visvesvaraya Technological University, Belagavi, during 2016-2020. It is certified that all the corrections/ suggestions indicated for internal assessment have been incorporated in the report & the project report has been approved as it satisfies the academic requirements.

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DECLARATION

We, **ROBIN BRIAN MIRANDA, RASHIKA NANDAKUMAR, PRERANA NIREEKSHA AMANNA & STEVENS KUNJUMMON CHACKO** hereby declare that the work titled **"NOISE REDUCTION OF LANDING GEAR USING ENHANCED MULTI-ELECTRODE DIELECTRIC BARRIER DISCHARGE PLASMA ACTUATOR"** embodied in this report has been carried out by us during 8th semester of BE degree at MVJCE Bangalore under the esteemed guidance of **Dr. P. ARUNAGIRI** (Assistant Professor, Department of Aeronautical Engineering, MVJ College of Engineering) affiliated to Visvesvaraya Technological University, BELAGAVI. The work embodied in this dissertation is original & it has as not been submitted in part of full for any other degree in any University.

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ACKNOWLEDGEMENT

We are extremely thankful to the principal **Dr. P. Mahabaleshwarappa** for sharing knowledge and assistance throughout our college period at MVJ College of Engineering, Bangalore.

In addition, we want to thank, **Dr. John Baruch** Head, Department of Aeronautical Engineering, MVJ College of Engineering for his valuable support and guidance. We are deeply touched by his transparent approach and whole hearted cooperation for execution of our report.

We express our deep sense of gratitude to our internal guide **Dr. P. Arunagiri** Assistant Professor Department of Aeronautical Engineering, MVJ College of Engineering for his valuable guidance at each and every steps of our report.

We would like to thank all the Professors at the Department of Aeronautical Engineering, MVJ College of Engineering, Bangalore for kind permission to carry out our project work and for support and guidance during project work.

This report certainly would not have been possible without the love and encouragement extended by our friends to take the opportunity to thank for their collective love and support.

ABSTRACT

With the development of low-noise aircraft engines, airframe noise now represents a major noise source during the commercial aircraft's approach to take-off or landing phase. Noise control efforts have therefore been extensively focused on the airframe noise problems in order to further reduce aircraft overall noise.

In this project, various control methods will be explored for noise reduction on airframe components especially the landing gear. We will be targeting major achievements in noise reduction with passive control methods such as brake fairings, wheel caps, undertray, leg-door filler, articulation link cover and then discuss the potential and control mechanism of some promising active flow control strategies for airframe noise reduction and thereby focusing mainly on Enhanced Multi-Electrode Dielectric Barrier Discharge (DBD) Plasma Actuator.

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CHAPTER 1

INTRODUCTION

1.1. INTRODUCTION TO AIRFRAME NOISE

Although aircraft, and especially their engines, have become significantly quieter in the last few decades, the overall levels of environmental annoyance are increasing as a result of the growth in aircraft traffic. European aircraft and engine manufacturers are facing increasing pressure to reduce aircraft noise levels; this pressure results from increased community expectations for an improved quality of life, and from the need to compensate for the expected growth in air traffic.

The continual development of quiet engines over the years has resulted in airframe noise levels becoming comparable with the engine noise at approach. Airframe noise is defined as the noise generated as a result of the aircraft moving through the air. The main airframe components that lead to airframe noise radiation are landing gears and high lift devices. Landing gear noise is one of the dominant noise sources from an aircraft during approach phase due to low engine rate setting. With the development of low-noise aircraft engine, airframe noise now represents a major noise source during the commercial aircraft’s approach to landing phase. Noise control efforts have therefore been extensively focused on the airframe noise problems in order to further reduce aircraft overall noise.

During an aircraft’s approach for landing what you're probably hearing is the sudden onset of the wind noise - when retracted the landing gear is streamlined, but when extended there is now an open cavity where the gear had been, and this open area in the fuselage creates a lot of wind noise from the suddenly turbulent airflow. Going from a relatively quiet state to a loud one suddenly gives the impression of a "bang." This is possibly enhanced by whatever noise is made by the up locks that hold the landing gear in place releasing, and perhaps some louder noise caused by the hydraulic motor; as the initial moments of the gear extension may be noisier than the steady-state noise of being fully extended. Also, the noise due to the turbulence will subside as the airspeed slows, and extending the landing gear often comes during a phase of flight when the airspeed is being reduced in preparation for the final approach segment and the subsequent landing.

1.2. PROBLEM STATEMENT

Due to the increase in civil air traffic coupled with a large population in the vicinity of major airports has resulted in increased community noise impact. Indeed, noise issues represent the primary environmental concern for our air transportation system.

The jet noise component of overall aircraft noise has been significantly reduced by the utilization of engines with high bypass ratio. During the landing phase, when engines are throttled down, airframe noise now represents a primary noise source. Therefore, our primary objective will be to reduce landing gear noise as aircraft noise will remain a constraint to the growth of the commercial air transportation system.

The current project is aimed to investigate the level of noise produced by a Conventional Landing Gear and to further reduce noise in a Landing Gear and other undercarriage subsystems using an Enhanced Multi-Electrode Dielectric Barrier Discharge (DBD) Plasma Actuators.

1.3. AIRCRAFT LANDING GEAR

Landing Gear system is one of the critical subsystems of an aircraft and is often configured along with the aircraft structure because of its substantial influence on the aircraft structural configuration itself. Landing gear detail design is taken up early in the aircraft design cycle due to its long product development cycle time. The need to design landing gear with minimum weight, minimum volume, reduced life cycle cost, and short development cycle time, poses many challenges to landing gear designers and practitioners. These challenges have to be met by employing advanced technologies, materials, analysis methods, processes and production methods. Various design and analysis tools have been developed over the years and new ones are still being developed.

The purpose of the landing gear in an aircraft is to provide a suspension system during taxi, take-off and landing. It is designed to absorb and dissipate the kinetic energy of landing impact, thereby reducing the impact loads transmitted to the airframe. The landing gear also facilitates braking of the aircraft using a wheel braking system and provides directional control of the aircraft on ground using a wheel steering system. It is often made retractable to minimize the aerodynamic drag on the aircraft while flying.

The landing gear design takes into account various requirements of strength, stability, stiffness, ground clearance, control and damping under all possible ground attitudes of the aircraft. These requirements are stipulated by the Airworthiness Regulations to meet operational requirements and safety. The landing gear should occupy minimum volume in order to reduce the stowage space requirement in the aircraft. Further, weight should be at minimum to increase the performance of the aircraft. The service life of the landing gears should be same as that of the aircraft.

A Landing Gear system comprises of many structural and system components. The structural components include Main fitting, Shock absorber, Bogie beam/ Trailing arm, Axle, Torque links, Drag/ Side braces, Retraction actuator, Down lock mechanism, Up lock, Wheel, Tire etc. The system components are Brake unit, Antiskid system, retraction system components.

Typical Main Landing Gear (MLG) is shown in Figure. The nose gear will have additional elements like steering actuator and steering mechanism.[1]

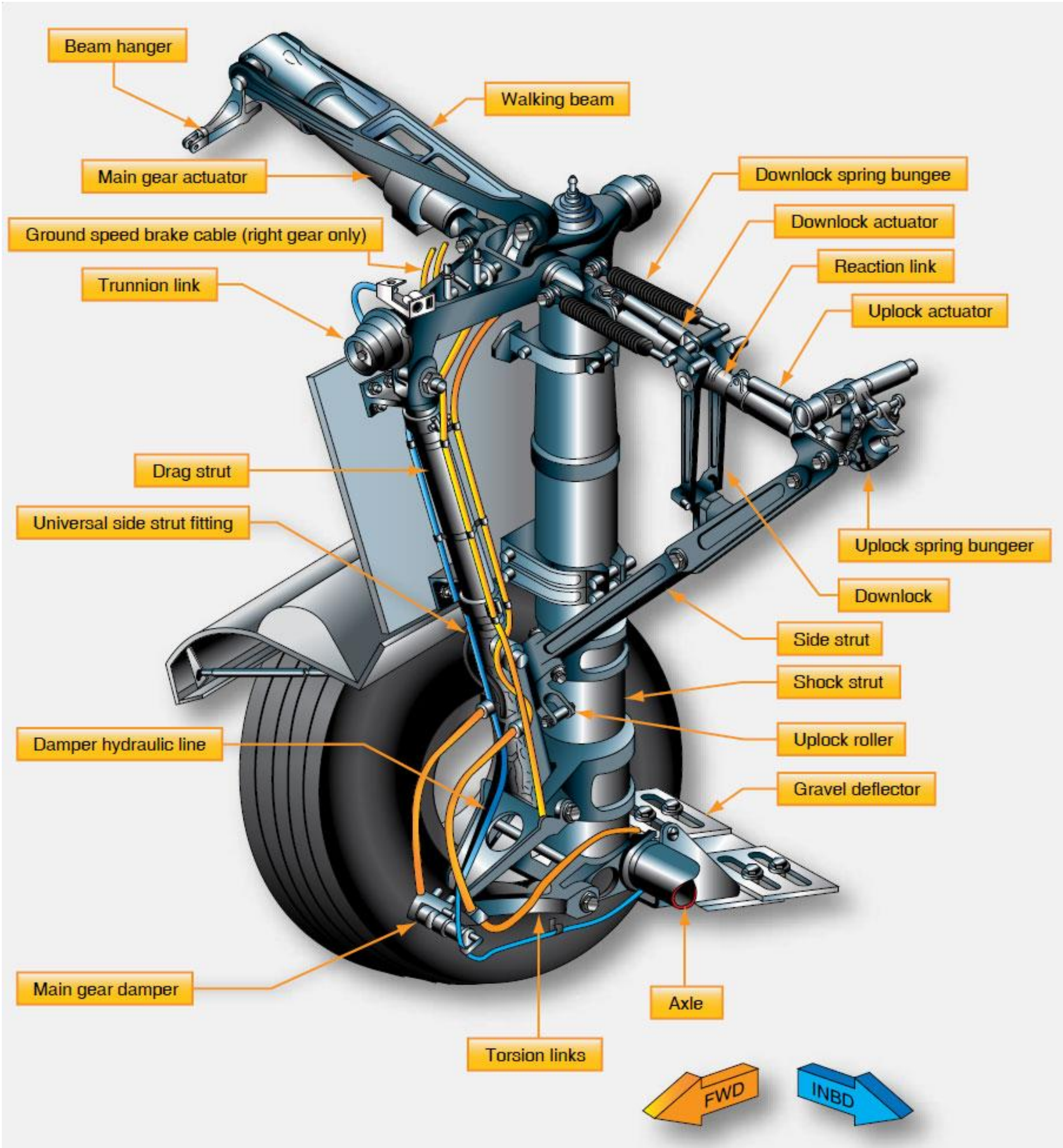

Fig 1.1. *Typical Main Landing Gear* [1]

1.4. AIRCRAFT LANDING GEAR TYPES

There are three basic arrangements of landing gears that are used: tail wheel-type landing gear (also known as conventional gear), tandem landing gear, and tricycle-type landing gear.

1.4.1. TAIL WHEEL TYPE LANDING GEAR

Tail wheel-type landing gear is also known as conventional gear because many early aircraft use this type of arrangement. The main gears are located forward of the center of gravity, causing the tail to require support from a third wheel assembly. A few early aircraft designs used a skid rather than a tail wheel. This helps slow the aircraft upon landing and provides directional stability. The resulting angle of the aircraft fuselage, when fitted with conventional gear, allows the use of a long propeller that compensates for older, underpowered engine design. The increased clearance of the forward fuselage offered by tail wheel-type landing gear is also advantageous when operating in and out of non-paved runways. Today, aircraft are manufactured with conventional gear for this reason and for the weight savings accompanying the relatively light tail wheel assembly.

The proliferation of hard surface runways has rendered the tail skid obsolete in favor of the tail wheel. Directional control is maintained through differential braking until the speed of the aircraft enables control with the rudder. A steerable tail wheel, connected by cables to the rudder or rudder pedals, is also a common design. Springs are incorporated for dampening.[1]

Fig 1.2. *Tail wheel configuration landing gear on a DC-3* [1]

1.4.2. TANDEM LANDING GEAR

Few aircraft are designed with tandem landing gear. As the name implies, this type of landing gear has the main gear and tail gear aligned on the longitudinal axis of the aircraft. Sailplanes commonly use tandem gear, although many only have one actual gear forward on the fuselage with a skid under the tail. A few military bombers, such as the B-47 and the B-52, have tandem gear, as does the U2 spy plane. The VTOL Harrier has tandem gear but uses small outrigger gear under the wings for support. Generally, placing the gear only under the fuselage facilitates the use of very flexible wings.[1]

Fig 1.3. *Tandem landing gear configuration on Boeing 777* [1]

1.4.3. TRICYCLE TYPE LANDING GEAR

The most commonly used landing gear arrangement is the tricycle-type landing gear. It is comprised of main gear and nose gear.

Tricycle-type landing gear is used on large and small aircraft due to the following benefits:

- Allows more forceful application of the brakes without nosing over when braking, which enables higher landing speeds.

- Provides better visibility from the flight deck, especially during landing and ground maneuvering.
- Prevents ground-looping of the aircraft. Since the aircraft center of gravity is forward of the main gear, forces acting on the center of gravity tend to keep the aircraft moving forward rather than looping, such as with a tail wheel-type landing gear.[1]

Fig 1.4. *Tricycle type landing gear on a Cessna 172* [1]

The nose gear of a few aircraft with tricycle-type landing gear is not controllable. It simply casters as steering is accomplished with differential braking during taxi. However, nearly all aircraft have steerable nose gear. On light aircraft, the nose gear is directed through mechanical linkage to the rudder pedals. Heavy aircraft typically utilize hydraulic power to steer the nose gear. Control is achieved through an independent tiller in the flight deck.

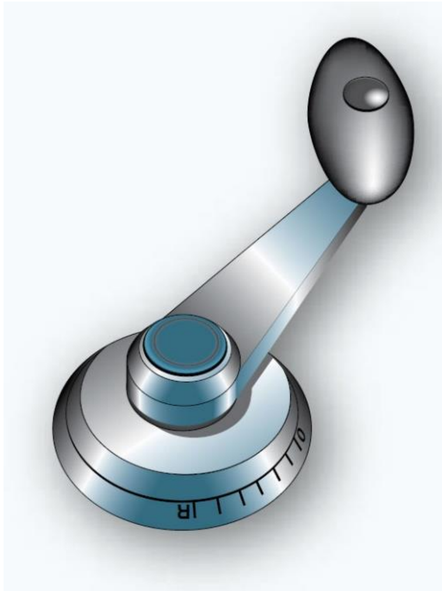

Fig 1.5. *A nose wheel steering tiller located on the flight deck* [1]

1.4.4. FIXED AND RETRACTABLE LANDING GEAR

Further classification of aircraft landing gear can be made into two categories: fixed and retractable. Many small, single-engine light aircraft have fixed landing gear, as do a few light twins. This means the gear is attached to the airframe and remains exposed to the slipstream as the aircraft is flown. As the speed of an aircraft increases, so does parasite drag. Mechanisms to retract and stow the landing gear to eliminate parasite drag add weight to the aircraft. On slow aircraft, the penalty of this added weight is not overcome by the reduction of drag, so fixed gear is used. As the speed of the aircraft increases, the drag caused by the landing gear becomes greater and a means to retract the gear to eliminate parasite drag is required, despite the weight of the mechanism.

A great deal of the parasite drag caused by light aircraft landing gear can be reduced by building gear as aerodynamically as possible and by adding fairings or wheel pants to streamline the airflow past the protruding assemblies. A small, smooth profile to the oncoming wind greatly reduces landing gear parasite drag. Figure 1.6 illustrates a Cessna aircraft landing gear used on many of the manufacturer’s light planes. The thin cross section of the spring steel struts combines with the fairings over the wheel and brake assemblies to raise performance of the fixed landing gear by keeping parasite drag to a minimum.[1]

Fig 1.6. *Main landing gear fairings on a Cessna 172* [1]

Retractable landing gears stow in fuselage or wing compartments while in flight. Once in these wheel wells, gears are out of the slipstream, they do not cause parasite drag. Most retractable gears have a close-fitting panel attached to them that fairs with the aircraft skin when the gear is fully retracted as shown in figure. Other aircrafts have separate doors that open, allowing the gear to enter or leave, and then close again.

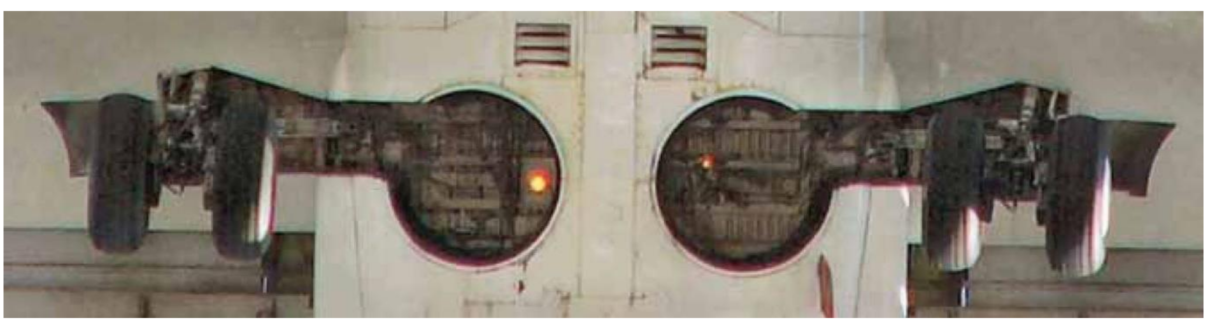

Fig 1.7. *The retractable gear of a Boeing 737 fair into recesses in the fuselage* [1]

NOTE: The parasite drag caused by extended landing gear can be used by the pilot to slow the aircraft. The extension and retraction of most landing gear is usually accomplished with hydraulics. Landing gear retraction systems are discussed below.

1.4.5. SHOCK ABSORBING AND NON-SHOCK ABSORBING LANDING GEAR

In addition to supporting the aircraft for taxi, the forces of impact on an aircraft during landing must be controlled by the landing gear. This is done in two ways:

- The shock energy is altered and transferred throughout the airframe at a different rate and time than the single strong pulse of impact.
- The shock is absorbed by converting the energy into heat energy.

1.4.5.1. Leaf Type Spring Gear

Many aircraft utilize flexible spring steel, aluminum, or composite struts that receive the impact of landing and return it to the airframe to dissipate at a rate that is not harmful. The gear flexes initially and forces are transferred as it returns to its original position. [Figure 1.8] The most common example of this type of non-shock absorbing landing gear are the thousands of single-engine Cessna aircraft that use it. Since these non-shock absorbing struts are made from steel, aluminum, or composite material, they are lighter in weight with greater flexibility and do not corrode. They transfer the impact forces of landing to the airframe at a non-damaging rate.

Fig 1.8. *Leaf Type Spring Landing Gear* [1]

1.4.5.2. Rigid

Before the development of curved spring steel landing struts, many early aircraft were designed with rigid, welded steel landing gear struts. Shock load transfer to the airframe is direct with this design. Use of pneumatic tires aids in softening the impact loads. [Figure 1.9] Modern aircraft that use skid-type landing gear make use of rigid landing gear with no significant ill effects. Rotorcraft, for example, typically experience low impact landings that are able to be directly absorbed by the airframe through the rigid gear (skids).

Fig 1.9. *Rigid steel landing gear is used on many early aircraft* [1]

1.4.5.3. Bungee Cord

The use of bungee cords on non-shock absorbing landing gear is common. The geometry of the gear allows the strut assembly to flex upon landing impact. Bungee cords are positioned between the rigid airframe structure and the flexing gear assembly to take up the loads and return them to the airframe at a non-damaging rate. The bungees are made of many individual small strands of elastic rubber that must be inspected for condition. Solid, donut-type rubber cushions are also used on some aircraft landing gear. [Figure 1.10]

Fig 1.10. *Piper Cub bungee cord landing gear transfer landing loads to the airframe* [1]

1.4.5.4. Shock Struts

True shock absorption occurs when the shock energy of landing impact is converted into heat energy, as in a shock strut landing gear. This is the most common method of landing shock dissipation in aviation. It is used on aircraft of all sizes. Shock struts are self-contained hydraulic units that support an aircraft while on the ground and protect the structure during landing. They must be inspected and serviced regularly to ensure proper operation.

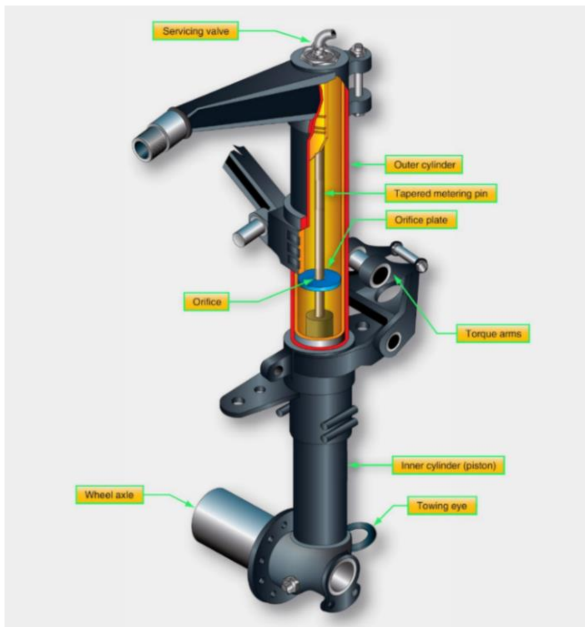

Fig 1.11. *Oleo-pneumatic shock struts used in Boeing 777's* [1]

For the purpose of our project we will be using Retractable Oleo-pneumatic shock strut type of Landing Gear as used in Boeing 777.

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CHAPTER 2

LITERATURE SURVEY

2.1. CONTROL STRATEGIES FOR AIRCRAFT AIRFRAME NOISE REDUCTION

Li, Y., Wang, X., & Zhang, D. [8] have explained in their thesis on "Control strategies for aircraft noise reduction" that with the development of low-noise aircraft engine, airframe noise now represents a massive noise source during the commercial aircraft's approach to landing phase. Noise control efforts have therefore been extensively focused on the airframe noise problems in order to further reduce aircraft overall noise. In this review, various control methods explored in the last decades for noise reduction on airframe components including high-lift devices and landing gears are summarized. We introduce recent major achievements in airframe noise reduction with passive control methods such as fairings, deceleration plates, splitter plates, acoustic liners, slat cover and side-edge replacements, and then discuss the potential and control mechanism of some promising active flow control strategies for airframe noise reduction, such as plasma technique and air blowing/suction devices. Based on the knowledge gained throughout the extensively noise control testing, a few design concepts on the landing gear, high-lift devices and whole aircraft are provided for advanced aircraft low-noise design. Finally, discussions and suggestions are given for future research on airframe noise reduction.

2.1.1. Airframe noise generation mechanism

One of the major airframe noise sources is landing gears. The noise generated by a landing gear is normally broadband in nature. Several noise sources have been identified on a typical landing gear configuration. The wheels and main struts are responsible for low frequency noise whilst the smaller details, such as the hoses and dressings, are responsible for the high frequency noise.

2.2. EXPERIMENTAL STUDY ON AIRCRAFT LANDING GEAR NOISE

Guo, Y., Yamamoto, K. J., & Stoker, R. W. [2] have conducted an experimental investigation on aircraft landing gear noise which is presented in their thesis "Experimental Study of Aircraft Landing Gear Noise". The study includes systematic testing and data analysis using a full-scale Boeing 737 landing gear.

The database covers a range of mean flow Mach numbers typical of landing conditions for commercial aircraft and various landing gear configurations, ranging from a fully dressed, complete gear to cleaner configurations involving only some parts of the complete gear. This enables the examination of noise radiation from various groups of the gear assembly and the derivation of functional dependencies of the radiated noise on the flow Mach number at various far-field directivity angles and on various gear geometry parameters. It is shown that the noise spectrum can be decomposed into three frequency components, namely, the low, mid and high-frequency components respectively, representing contributions from the wheels, the main struts and the small details such as hoses, wires, cutouts, and steps.

It is found that these different frequency components have different dependencies on flow parameters and gear geometry. Based on the spectral decomposition in the three frequency domains, normalized spectra are derived for all three components and a model for the overall sound levels is developed as a function of flow and geometry parameters.

2.3. ACTIVE FLOW CONTROL BY USING PLASMA ACTUATORS

Neretti, G.[3] has written in his thesis “Active Flow Control by Using Plasma Actuators” about the active flow control strategies using Plasma Actuators which allows it to directly manipulate the flow-field around a surface only when it is effectively requested, due to which the active flow control techniques has recently received an increasing attention. Aerodynamic plasma actuators supplied by a dielectric barrier discharge (DBD) can be used for this purpose. Usually, sinusoidal voltages in the range 5–50 kV peak and frequencies between 1 and 100 kHz are utilized to ignite this plasma typology. The surface discharge produced by these devices is able to tangentially accelerate the flow field by means of the electrohydrodynamic (EHD) interaction. DBDs generate non-thermal plasmas characterized by low input energies and limited temperature increments.

Plasma actuators can be easily designed by following the shape of the aerodynamic body and can be used over heat-sensitive surfaces. These aerodynamic devices have demonstrated to produce boundary layer modifications with induced speeds up to 10 m/s. Their use over airfoils, flaps, and blades have shown the possibility to delay the transition between laminar to turbulent regime, to prevent flow separation enhancing lift and reducing drag. Moreover, the adoption of these actuators over landing gears and trailing edges may induce a noise reduction effect. Dielectric materials, electrodes configuration, and supplying waveforms are most relevant parameters to be considered to enhance actuator performance. On a parallel plane, on/off actuation strategy is a key point in the use of these devices when utilized over aerodynamic surfaces impinged within an external flow.

2.4. PLASMA ACTUATORS FOR LANDING GEAR NOISE REDUCTION

Thomas, F., Kozlov, A., & Corke, T. [4] have explained about the use of Plasma Actuators for the reduction of noise in Landing Gear in their thesis “Plasma Actuators for Landing Gear Noise Reduction”. A primary component of airframe noise on both takeoff and landing approach is due to the landing gear. The inherent bluff body characteristics of the landing gear give rise to large-scale flow separation that results in noise production through unsteady wake flow and large-scale vortex instability and deformation. In this paper the use of single dielectric barrier discharge plasma actuator technology for landing gear noise control is explored. Proof-of-concept experiments that use plasma actuators to create an effective “plasma fairing” which minimizes flow separation over the gear are presented.

Most active flow-control devices currently available control flow by changing the shape of the device using some mechanical moving parts (e.g., leading edge slats or trailing edge flaps to control the flow around an airfoil). These mechanical flow-control devices can modify the flow and enhance the performance of fluid machinery.

Active flow control techniques using the electrohydrodynamic (EHD) effect are one of the most promising candidates for next-generation flow-control devices, owing to their advantages over existing flow control devices i.e., because they have no moving parts and offer high-speed responsiveness. The EHD effect induces an ionic wind, which modifies the flow field around the fluid machinery and generates propulsive force. Active flow-control devices that use EHD force are referred to as plasma actuators.

2.5. SUCCESSIVELY ACCELERATED IONIC WIND WITH INTEGRATED DBD PLASMA ACTUATOR FOR LOW-VOLTAGE OPERATION

Sato, S., Furukawa, H., Komuro, A., Takahashi, M., & Ohnishi, N. [5] have explained in their thesis stated above about successively accelerating the ionic wind with integrated dielectric-barrier-discharge plasma actuator for low voltage operation. Electrohydrodynamic (EHD) force is used for active control of fluid motion and for the generation of propulsive thrust by inducing ionic wind with no moving parts.

We propose a method of successively generating and accelerating ionic wind induced by surface dielectric-barrier-discharge (DBD), referred to as a DBD plasma actuator with multiple electrodes. A conventional method fails to generate unidirectional ionic wind, due to the generation of a counter ionic-wind with the multiple electrodes DBD plasma actuator.

However, unidirectional ionic wind can be obtained by designing an applied voltage waveform and electrode arrangement suitable for the unidirectional EHD force generation. Our results demonstrate that mutually enhanced EHD force is generated by using the multiple electrodes without generating counter ionic-wind and highlights the importance of controlling the dielectric surface charge to generate the strong ionic wind. the proposed method can induce strong ionic wind without a high-voltage power supply, which is typically expensive and heavy, and is suitable for equipping small unmanned aerial vehicles with a DBD plasma actuator for a drastic improvement in the aerodynamic performance.

2.6. MODELLING OF DIELECTRIC BARRIER DISCHARGE PLASMA ACTUATORS FOR DIRECT NUMERICAL SIMULATIONS

Brauner, T., Laizet, S., Benard, N., & Moreau, E. [6] have explained in their thesis stated above about the modeling of dielectric plasma actuators for direct numerical simulations. In recent years the development of devices known as plasma actuators has advanced the promise of controlling flows in new ways that increase lift, reduce drag and improve aerodynamic efficiencies; advances that may lead to safer, more efficient and quieter air-craft. The large number of parameters (location of the actuator, orientation, size, relative placement of the embedded and exposed electrodes, materials, applied voltage, frequency) affecting the performance of plasma actuators makes their development, testing and optimization a very complicated task. Several approaches have been proposed for developing numerical models for plasma actuators.

The discharge can be modelled by physics-based kinetic methods based on first principles, by semi-empirical phenomenological approaches and by PIV-based methods where the discharge is replaced by a steady-state body force. The latter approach receives a recent interest for its easy implementation in RANS and U-RANS solvers. Here, a forcing term extracted from experiments is implemented into our high-order Navier-Stokes solver (DNS) in order to evaluate its robustness and ability to mimic the effects of a surface dielectric barrier discharge. This experimental forcing term is compared to the numerical forcing term developed by Suzen & Huang with an emphasis on the importance of the wall-normal component of each model.

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CHAPTER 3

DESIGNING OF NOSE LANDING GEAR USING CATIA-V5

3.1. NOSE LANDING GEAR DESIGN PROCESS USING CATIA-V5

The design of the nose landing gear is obtained by using the required commands in CATIA.

Firstly, the reference nose landing gear design is chosen from Boeing – 777.

In designing this model, we used

- Part Design
- Generative Shape Design
- Assembly

The following screenshots represents the steps in this design process.

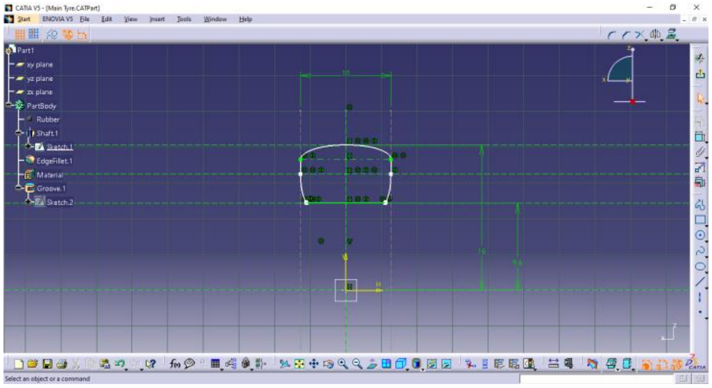

Fig 3.1. *2D Sketch of main tyre*

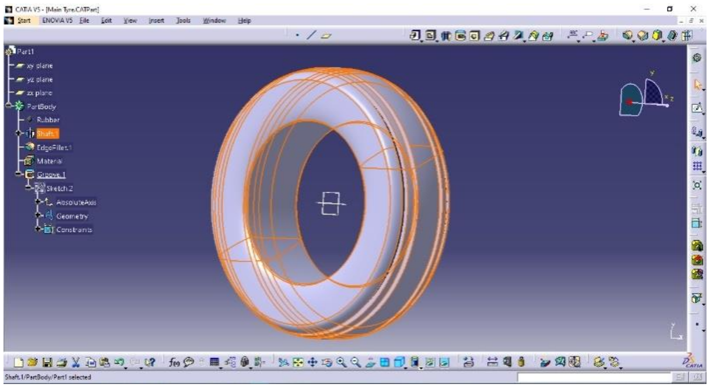

Fig 3.2. *Main tyre revolved*

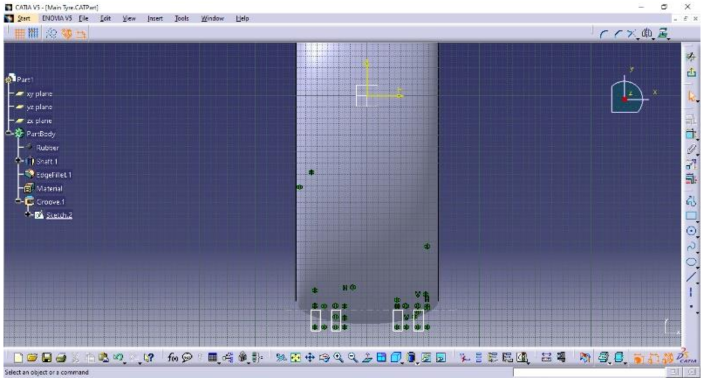

Fig 3.3. *Tyre thread sketch*

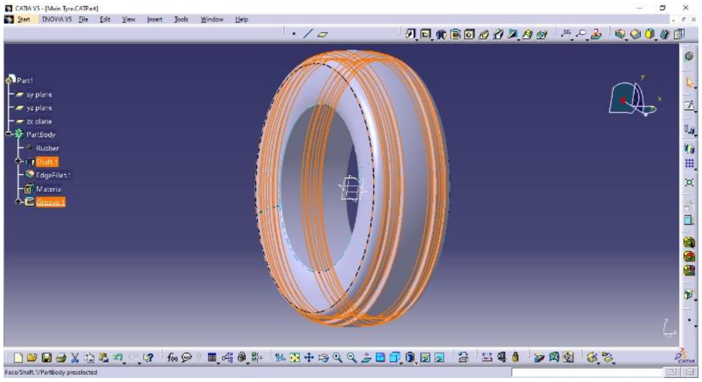

Fig 3.4. *Tyre thread grooving*

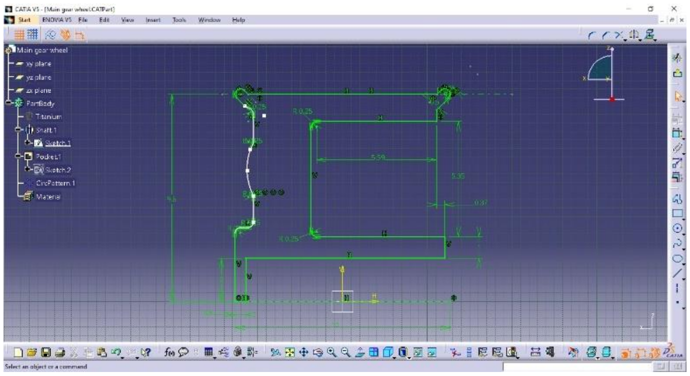

Fig 3.5. *2D sketch of main gear wheel rim*

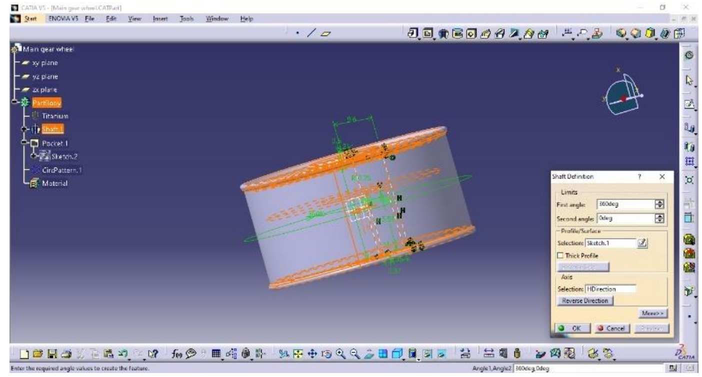

Fig 3.6. *Revolved main gear wheel rim*

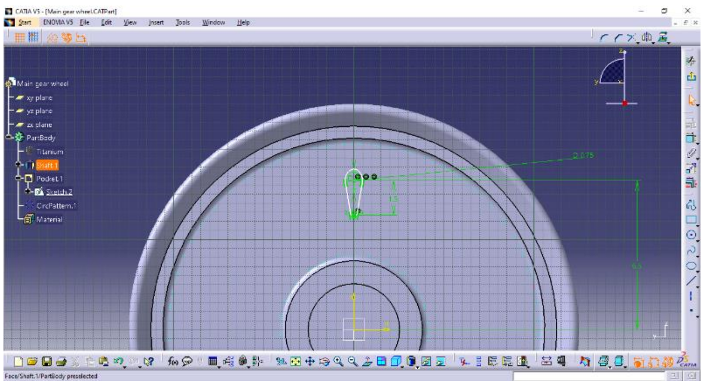

Fig 3.7. *2D sketch of rim hole design*

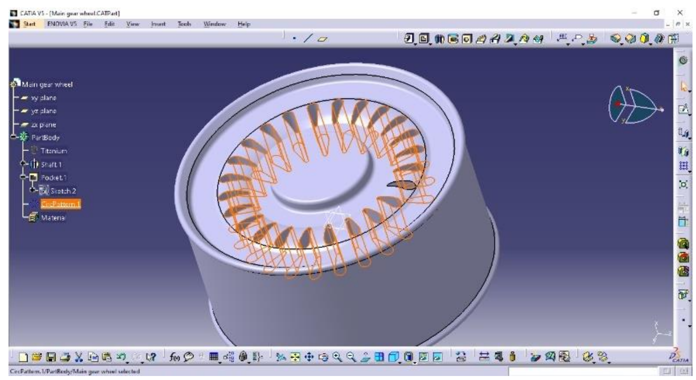

Fig 3.8. *Extruded cut and circular pattern*

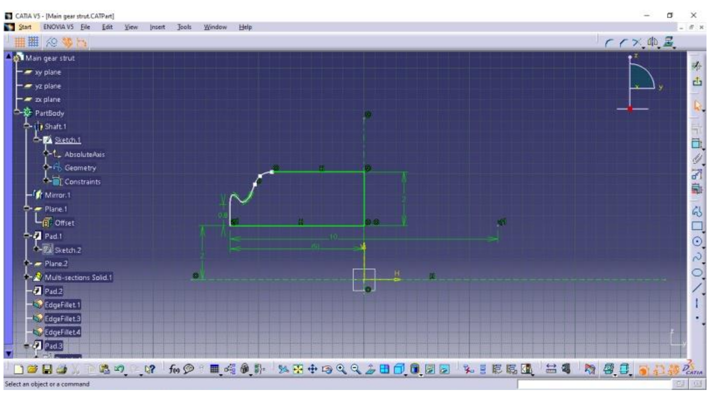

Fig 3.9. *2D sketch of main gear strut hub*

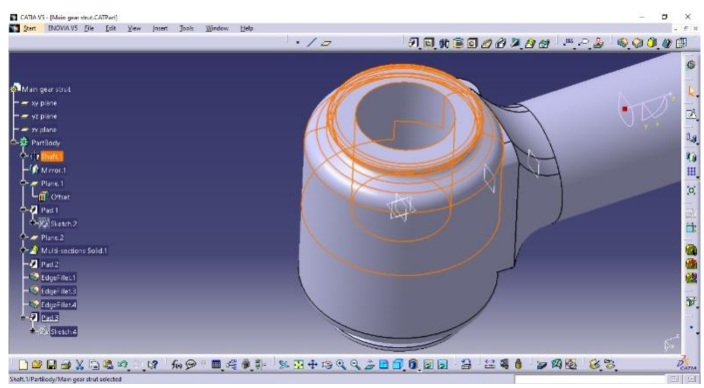

Fig 3.10. *Extruded hub*

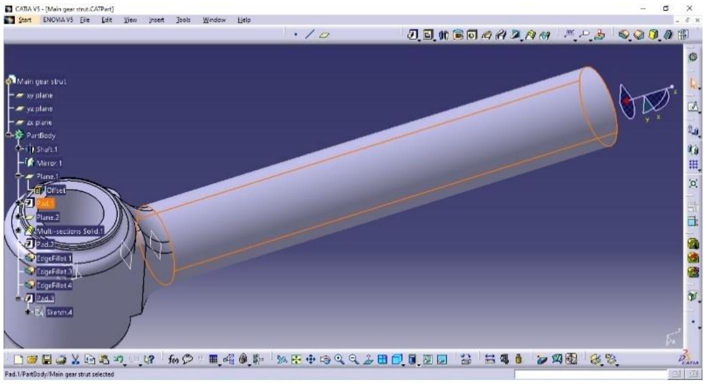

Fig 3.11. *Extruding the strut*

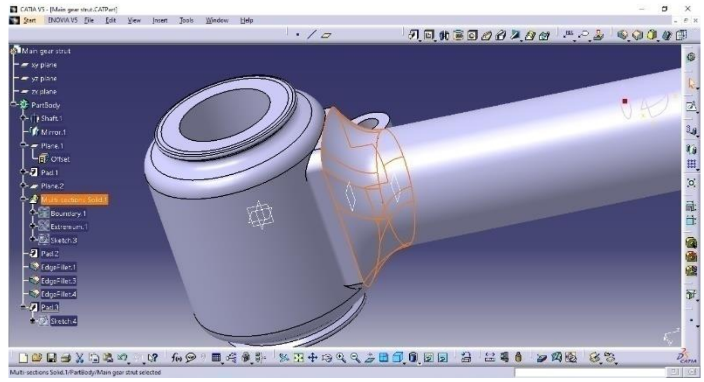

Fig 3.12. *Using multi-section command to loft the two parts together*

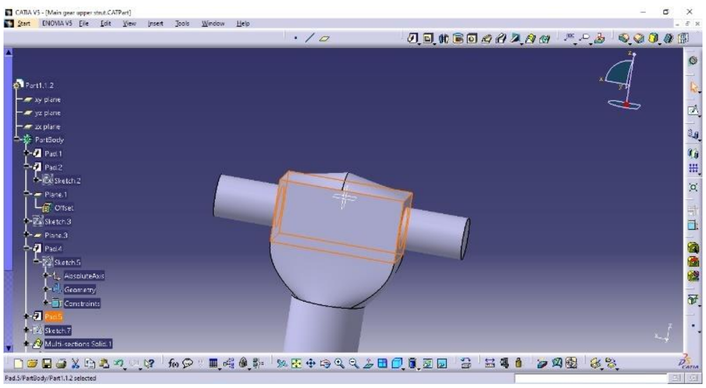

Fig 3.13. *Padding the upper connector and pins*

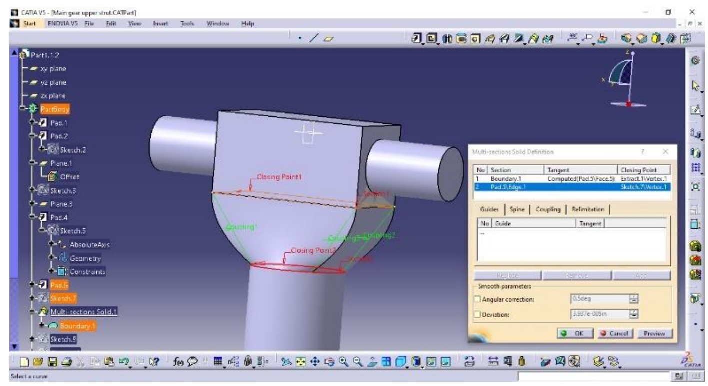

Fig 3.14. *Lofting the two parts together*

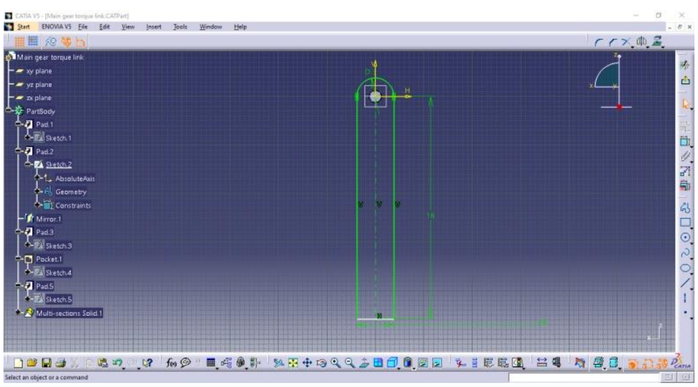

Fig 3.15. *2D sketch of main torque link*

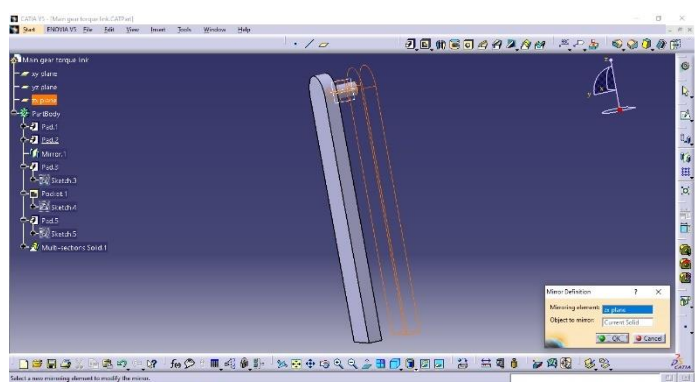

Fig 3.16. *Extruding the sketch and mirroring it*

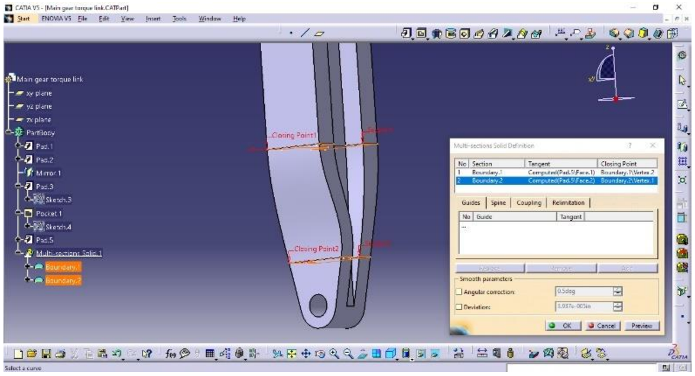

Fig 3.17. *Using multi-section solid to connect*

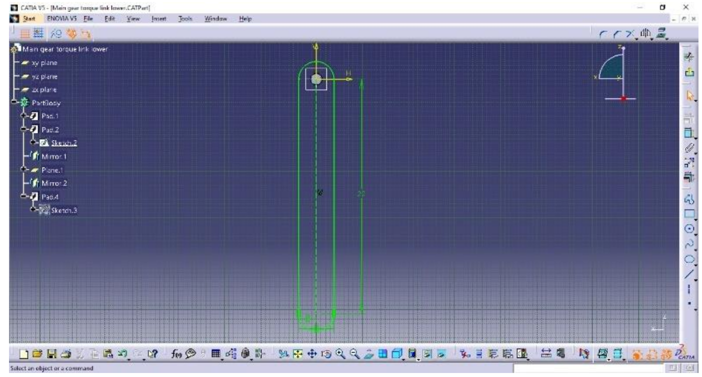

Fig 3.18. *2D sketch of lower torque link*

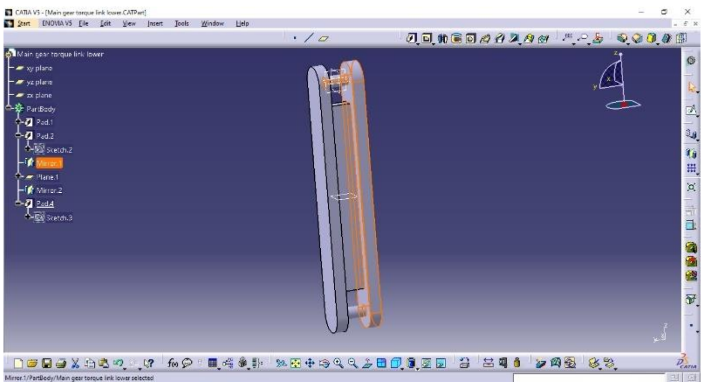

Fig 3.19. *Padding and mirroring the sketch*

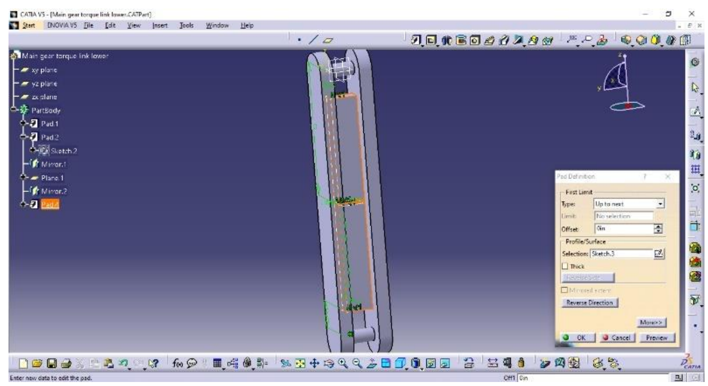

Fig 3.20. *Extruding the pins & the mid plate*

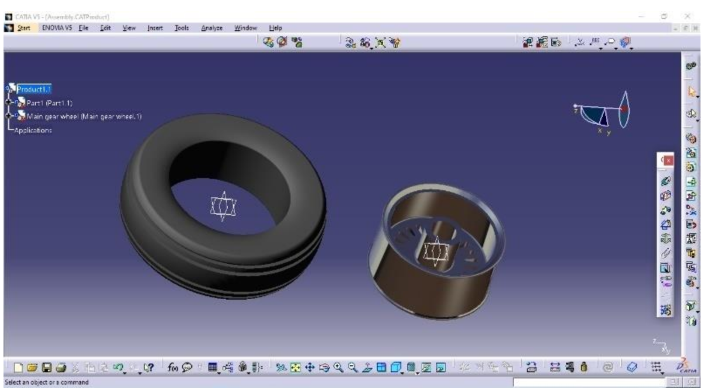

Fig 3.21. *Importing the existing components*

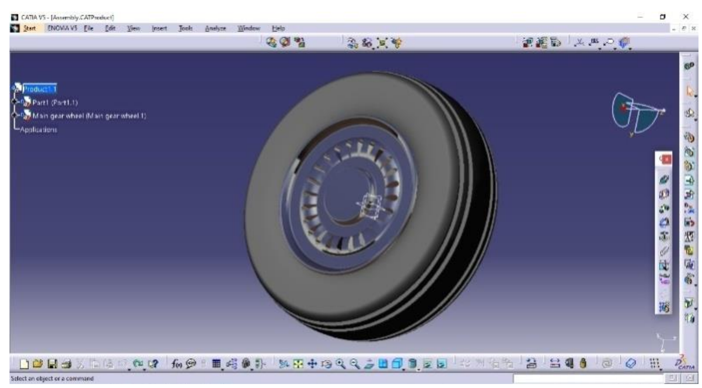

Fig 3.22. *Sub-Assembly of the main tyre*

Fig 3.23. *Exploded view of the main assembly*

Fig 3.24. *Final Assembly of the nose landing gear*

Fig 3.25. *Rendered view of the Nose Landing Gear of Boeing -777*

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CHAPTER 4

AIRCRAFT NOISE

4.1. AIRCRAFT NOISE AND ITS EFFECTS

Noise is defined as “unwanted sound.” Aircraft noise is one, if not the most harmful environmental effect of aviation. It can cause community annoyance, disrupt sleep, adversely affect the academic performance of children, and could increase the chance for cardiovascular disease of people living near the airports. In some airports, noise constraints affect the air traffic growth.[7]

Advances in jet engine technology over the past 5 decades have helped to achieve substantial reductions in aircraft noise compared to early jet airliners. However, additional efforts to develop quieter aircrafts are still necessary because of the prospective increase in the number of takeoffs and landings as a result of the growing demand for air travel. Also reducing the engine noise further may come at the cost of its operation and efficiency. One amongst the challenges to realizing quieter aircraft is reducing airframe noise, which is aerodynamic noise generated mainly by high-lift devices and landing gear. For modern aircraft with advanced very high-bypass ratio engines, this airframe noise is one of the dominant sources of aircraft noise during approach and landing.

With the development of low-noise aircraft engine, airframe noise currently represents a major noise source during the commercial aircraft’s approach to landing phase. Noise control efforts have thus been extensively focused on the airframe noise issues to further reduce aircraft overall noise.[8]

In this review, numerous control strategies explored within the last few decades for noise reduction on airframe components including high-lift devices and landing gears are summarized. We introduce recent major achievements in airframe noise reduction with passive control methods such as fairings, deceleration plates, splitter plates, acoustic liners, slat cover and side-edge replacements, and then discuss the potential and control mechanism of some promising active flow control strategies for airframe noise reduction, such as plasma technique and air blowing/suction devices. Based on the knowledge gained throughout the extensive noise control testing, a few design concepts on the landing gear are provided for advanced aircraft low-noise design. Finally, discussions and suggestions are given for future research on airframe noise reduction.[8]

4.2 CAUSES OF AIRCRAFT NOISE

The noise from a plane is caused by two things: one is from the air going over its body (or 'airframe') and the other is from its engines.

4.2.1 AIRFRAME NOISE

A moving aircraft causes friction and turbulence, which triggers sound waves. Generally, the faster the aircraft is flying, the more turbulence and friction will occur. When the aircraft's landing gear and flaps are used, more noise is made because more resistance is being created. The amount of noise this creates can vary according to the way the plane is flown, even for identical aircraft.

Another major source of airframe noise is the high-lift devices, including leading-edge slats and trailing-edge flaps. Other high-lift-related noise generating devices include spoilers if deployed during a steep approach operation. Although the spoiler noise may be subjectively important, it has little impacts for airworthiness and thus does not receive much attention for the noise research community by now and will not be addressed in this paper.[9]

4.2.2. ENGINE NOISE

Engine noise is caused by the sound of moving parts, and by the air coming out of the engine at high speed and interacting with still air, creating friction. Modern bypass engines, which introduce a layer of moderately fast-moving cold air between the hot exhaust and the still air, are quieter than early jet engines, which didn't use this technology.[9]

4.3. ILLUSTRATION OF NOISE SOURCES ON AN AIRCRAFT AT 1kHz-f

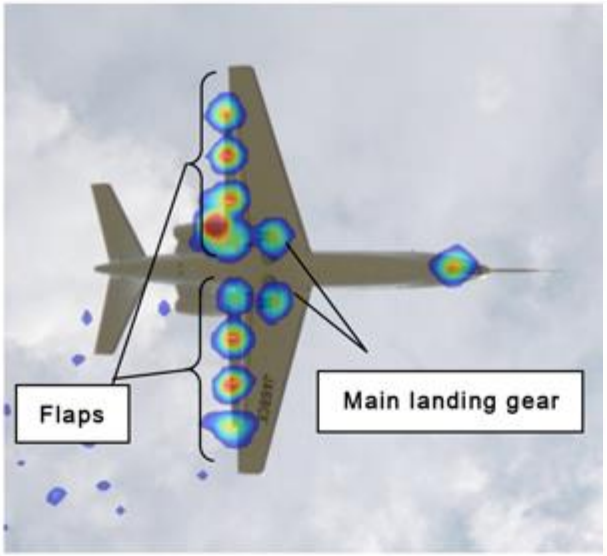

Fig 4.1. *Without noise reduction design (baseline aircraft)*

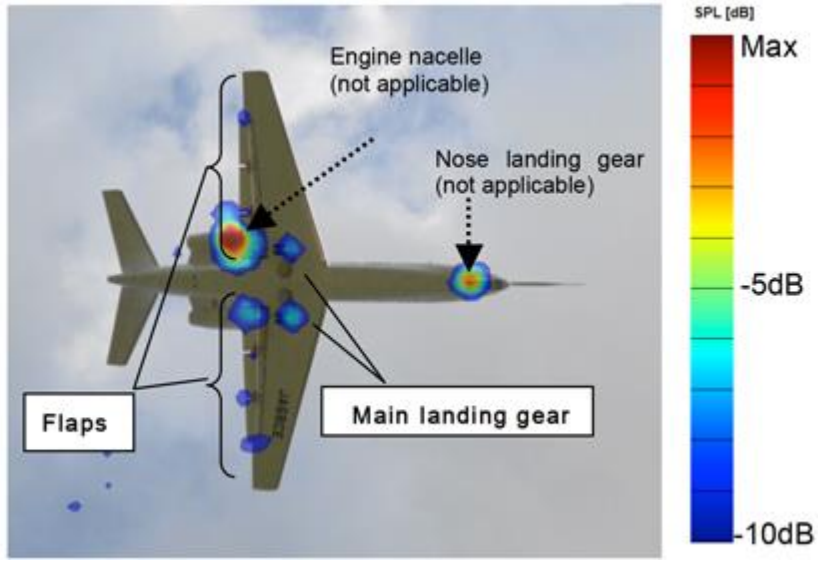

Fig 4.2. *With noise reduction design applied to the flaps and main landing gear (modified aircraft)*

The above figures show the main sources of noise on the aircraft (*JAXA's research aircraft 'Hisho' based on Cessna Model 680*) when subjected to 1Khz frequency band and 1/3 octave band noise spectra. It clearly shows that the main landing gear and flaps modification successfully reduces the landing gear noise at that frequency when compared to the baseline aircraft, while the un-modified engine nacelle and nose landing gear produce the same level of noise.[10]

4.4. ICAO CURRENT INITIATIVES ON AIRCRAFT NOISE

Limiting or reducing the number of people affected by significant aircraft noise is, therefore, one of ICAO's main priorities and one of the Organization's key environmental goals. Continuous work is being conducted by ICAO to reduce aircraft noise. This work includes, among several topics, investigations into emerging noise reduction technologies, noise impacts from new aircraft concepts (e.g. Unmanned Air Vehicles), and the development of SARPs for future supersonic airplanes. ICAO is also working on the environmental aspects of airport land-use planning, and good practices on airport community engagement.[11]

4.5. AIRCRAFT LANDING GEAR NOISE

Aircraft landing gear noise has been recognized in recent years as one of the major contributors to the total aircraft noise, especially at landing conditions. Its prediction, however, remains one of the most difficult challenges in aeroacoustics because of the complexity of the gear geometry and the surrounding flow field. Though significant progress has been made in computational power and numerical simulation methodologies in recent years, solving the unsteady flow field around a practical landing gear assembly and hence, computing the noise field, is still not a practical task. This is very likely to remain so for many years to come, and our best hope to develop useful prediction tools for the foreseeable future probably lies in empirical approaches and other approximate methods.[12]

One of the major airframe noise sources is the landing gears. The noise generated by a landing gear is normally broadband in nature. Several noise sources have been identified on a typical landing gear configuration. The wheels and main struts are responsible for low-frequency noise whilst the smaller details, such as the hoses and dressings, are responsible for the high-frequency noise. Some studies have shown tonal noise due to cavity resonances from tube-type pins in various joints linking gear components, tire treads and hinge-leg door configurations. It seems that this tonal noise is dependent on inflow velocity, turbulence and flow direction, so it is impossible to predict whether these will manifest themselves during the approach of an aircraft. It should be noted, however, that there is little experimental evidence that vortex shedding-related tone noise is a major problem for current landing gear architectures.[12]

On the other hand, the landing gear broadband noise is normally generated by the turbulence flow separation off the bluff-body components and the subsequent interaction of such turbulent wake flows with downstream located gear elements.[12]

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CHAPTER 5

FLOW CONTROL TECHNIQUES

5.1. PASSIVE FLOW CONTROL TECHNIQUES

5.1.1 THE FAIRING CONCEPT

An effective way to minimize the landing gear noise is to cover the gear components behind a fairing. In one of the papers published Dobrzynski [13], it was demonstrated that a total noise reduction of 10dB could be achieved if the entire gear structure was covered by one fairing to appear as a streamlined body. Although a faired landing gear generates considerably less noise than the corresponding unmodified gear, the need to access the gear for maintenance and stow the gear in cruise makes passive separation control via fairings impractical in case of commercial passenger jets.

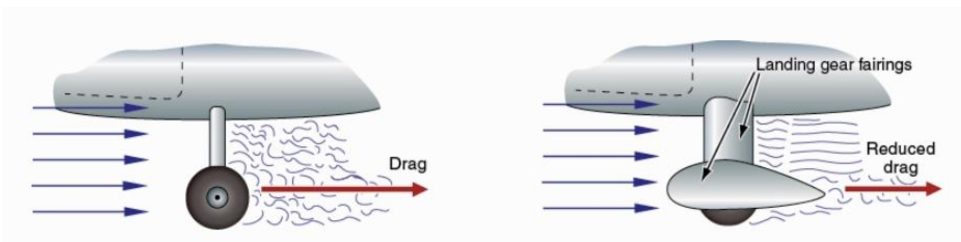

Fig 5.1. *The fairing causes streamlining which reduces form drag*

5.2. ACTIVE FLOW CONTROL TECHNIQUES

5.2.1. INTRODUCTION TO ACTIVE FLOW CONTROL TECHNIQUES

The interest in flow-control in the fluid dynamic domain has shown a quick growth in the last 15 years [15] due to the rapid development of sensors and actuators technologies. Even though passive control techniques are still attractive, since they do not require energy input, active control strategies have recently received more attention since they can be used selectively and can be operated only when it is effectively requested. Among different active techniques, plasma aerodynamic actuators are attractive because they present high dynamic responses due to the absence of moving parts, and are characterized by low weight, are easy to build, are backwards compatible with existing aerodynamic surfaces, and generate negligible aerodynamics interferences when they are switched off. When actuated, they can significantly modify the status of the boundary layer developing on the body surfaces. For this reason, they have been extensively studied for aeronautical applications to prevent flow separation enhancing lift and reducing drag. [16,17]

More recently, they have been used also to control friction drag by delaying transition [18,19] or by oscillating the flow in a spanwise direction [20] and to control global instabilities of the flow. Due to these characteristics, the potential of plasma actuators has been extended to many other applications like for instance tip clearance flow control of turbines, and wind turbine blades and holder. [21,22]

5.2.2. PLASMA ACTUATORS

A recent development in the field of aerodynamic flow control is that of plasma actuators which were first developed in the late 60s by Velkoff and Ketchman. Only in recent years, however (the mid-’90s until today) did plasma actuators undergo considerable research and development; thus, the first commercial applications are yet to come into realization.

The basic principle of plasma actuators is the creation of an electric field between two electrodes by applying a high voltage between them. The electric field that is consequently created induces a 2D wall jet close to the wall surface. This is possibly caused by the impact forces of the colliding plasma ions and the air molecules and particles in the actuator region. The plasma-induced jet adds momentum to the boundary layer and triggers flow instabilities thus modifying the boundary layer properties. The main parameters influencing the behaviour of the plasma actuators are:

- Electrode Shape
- Electrode Size
- Dielectric Gap
- Electrode Distances
- Voltage

- Frequency (for AC actuators)
- Waveform (for AC actuators)
- Nature of dielectric wall
- Free-stream flow velocity
- Humidity

Active flow control has recently received increasing attention since it allows to directly manipulate the flow-field around a surface only when it is effectively requested. Aerodynamic plasma actuators supplied by a dielectric barrier discharge (DBD) can be used for this purpose. Usually, sinusoidal voltages in the range 5–50 kV peak and frequencies between 1 and 100 kHz are utilized to ignite this plasma typology. The surface discharge produced by these devices can tangentially accelerate the flow field through the electrohydrodynamic (EHD) interaction. DBDs generate non-thermal plasmas characterized by low input energies and limited temperature increments.[14]

Plasma actuators can be easily designed by following the shape of the aerodynamic body and can be used over heat-sensitive surfaces. These aerodynamic devices have demonstrated to produce boundary layer modifications with induced speeds up to 10 m/s. Their use over airfoils, flaps, and blades have shown the possibility to delay the transition between laminar to the turbulent regime, to prevent flow separation enhancing lift and reducing drag. Moreover, the adoption of these actuators over landing gears and trailing edges may induce a noise reduction effect. Dielectric materials, electrodes configuration, and supplying waveforms are most relevant parameters to be considered to enhance actuator performance. On a parallel plane, on/off actuation strategy is a key point in the use of these devices when utilized over aerodynamic surfaces impinged within an external flow.[14]

Types of Plasma Actuators

Currently, there are several types of plasma actuators under development and each type has its advantages and disadvantages which are described in further detail in the following paragraphs.

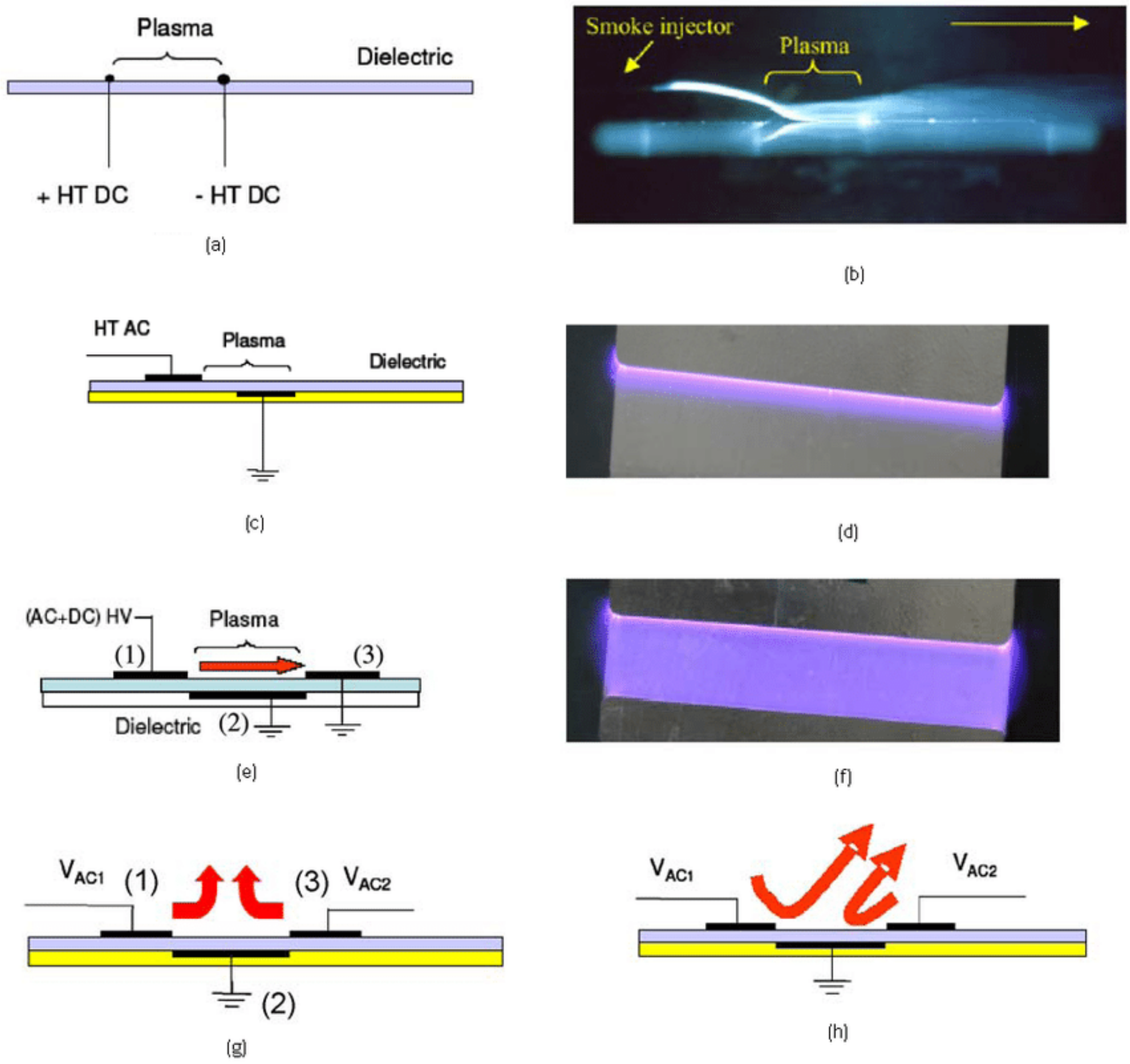

Fig 5.2. *Schematics and images of the different plasma actuator types. a) Schematic of a DC corona discharge actuator, b) Visualization of the effect of the actuator (a), c) AC Barrier discharge actuator, d) Visualization of the effect of the actuator (c), e) Three electrode discharge actuator, f) Visualization of the effect of the actuator (e), g) Wall jet actuator with balanced electrode voltage, h) Wall jet actuator with imbalanced electrode voltage* [61]

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CHAPTER 6

DBD PLASMA ACTUATOR

6.1. BASIC WORKING PRINCIPLE OF A DBD PLASMA ACTUATOR

Plasma aerodynamic actuators are based on the electrohydrodynamic (EHD) interaction generated by the so-called ionic wind. High electric fields can locally ionize the air. The produced heavy-charged species are accelerated by the applied electric field and, by means of collisions, they can yield momentum to the surrounding neutral gas. The force *f* per unit volume within the discharge can be yield by the following expression:

$$f = (n_i - n_e) \bar{E}$$

where E is the electric field and n_i and n_e are ion and electron number density respectively.

First studies were conducted by using direct current (DC) corona discharges.[23] Main disadvantages of these actuators are low flexibility in electrodes configuration and in the supply system, and the transition into spark regime producing irreversible damage of the actuator itself.

Nowadays, the largest part of plasma aerodynamic actuators is based on the surface dielectric barrier discharge (SDBD). This typology of discharge allows us to use several electrode geometries, different supply voltage waveforms, and it also prevents the transition into the arc regime due to the presence of the dielectric material. DBD actuators generate non-thermal plasmas [24], limiting power consumption and allowing their use over heat-sensitive surfaces.

A classical DBD aerodynamic plasma actuator is constituted by a dissymmetric electrode pair separated by a dielectric slab [Figure 6.1]. When high voltages (typically 10–50 kV at 1–100 kHz) are applied to the electrodes, a surface discharge is produced [Figure 6.2]. Discharge appears to be macroscopically homogeneous to the unaided eye, but it is constituted by a sequence of micro-discharges [25] lasting typically for tens of nanoseconds with a repetition rate of several hundreds of megahertz. Fast ignition and quenching of the discharge can be inferred by voltage-current time behaviour reported in Figure 6.3.

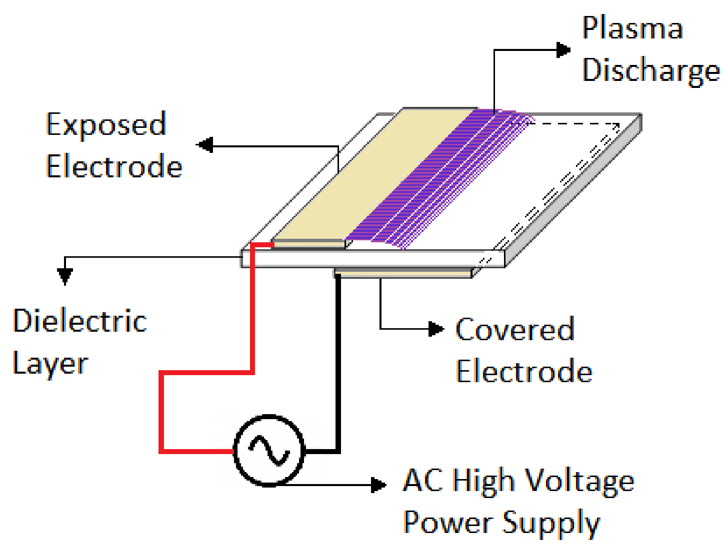


Fig 6.1. *Schematic of a SDBD Plasma Actuator for flow control* [68]

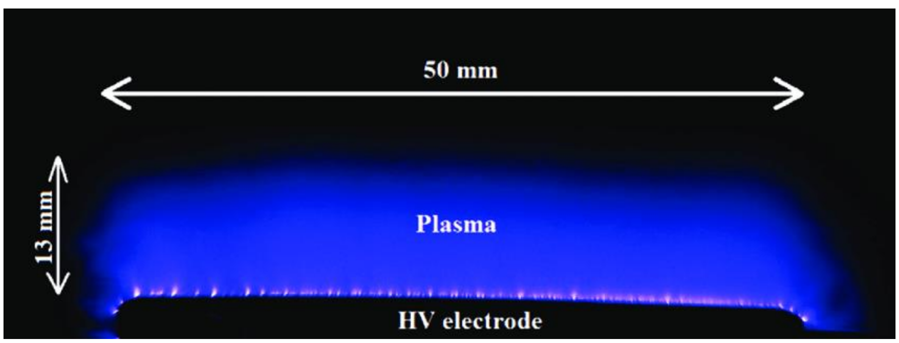


Fig 6.2. *Top view image of SDBD discharge* [32]

The presence of the plasma and the particular electrode configuration induces a jet tangential to the actuator wall. These jets can modify the aerodynamic boundary layer, increasing flow momentum, at least in the near-wall region above the surface. A large number of experimental [26–37] and numerical [38–59] works have been done in the last decade to understand basic physical phenomena involved in the EHD interaction.

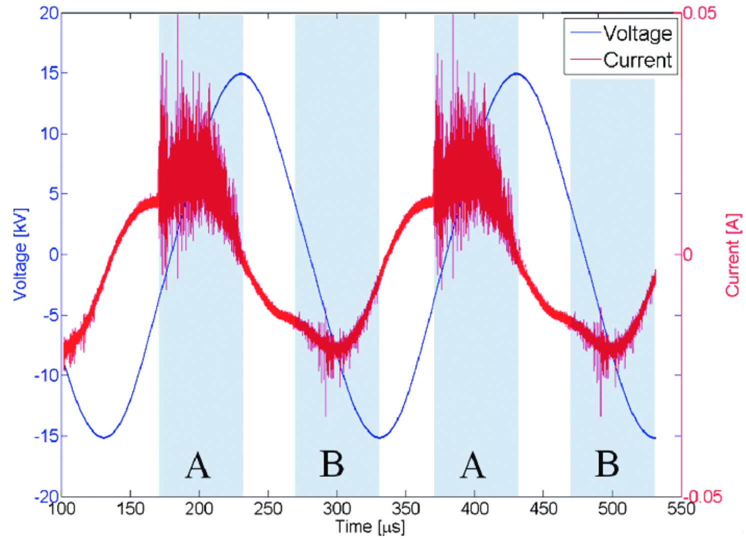


Fig 6.3. *Voltage-current time behaviour of a SDBD* [33]

A typical velocity profile induced by the tangential wall jet is shown in [Figure 6.4]. Measurements have been carried out by a glass Pitot tube positioned 2 mm downstream to the plasma extension and moved parallel to the actuator surface. Maximum velocity of about 5–6 m/s is usually reached. [20]

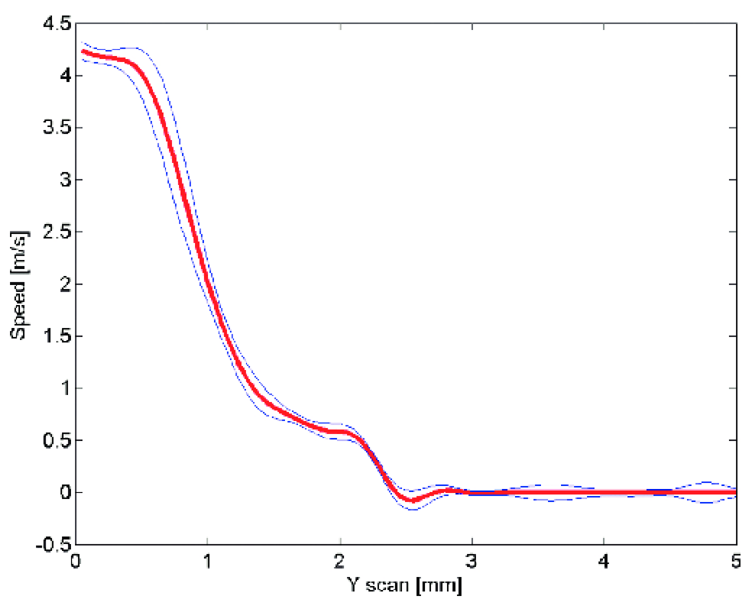


Fig 6.4. *Typical pitot velocity profile of a DBD plasma actuator (red line). Blue lines represent standard deviation* [66]

6.2. FLOW MANIPULATION OF DBD PLASMA ACTUATORS

In the last decade, DBD aerodynamic actuators have been extensively studied in the active flow control domain. Position and actuation strategies of these devices are key points in the effectiveness of flow control over aerodynamic surfaces. Many studies have been accomplished over airfoils, diffusers, and wind and turbine blades, in order to enhance fluid-dynamic performance. Moreover, the adoption of these devices over landing gears and trailing edge surfaces have demonstrated the possibility to obtain a noise reduction effect. DBD actuators can be usually positioned over aerodynamic surfaces in spanwise and streamwise directions. In the former, the induced body force is in the same direction as the incoming flow. In the latter, induced thrust is perpendicular to the free stream direction. In this case, the composition of these two flows produces vorticities propagating in the downstream direction.[14]

In [Figure 6.5], a NACA 0015 airfoil equipped with four spanwise DBD plasma actuators is shown. The four plasma regions are visible in the figure as bluish strips. In this configuration, actuators produce a tangential wall jet directed downstream. The effect of this plasma device in the recovery of stall condition is depicted in [Figure 6.6], where smoke visualization is reported. Experiments show how the most effective actuator is the one positioned on the airfoil leading edge, just before the region where separation occurs.[14]

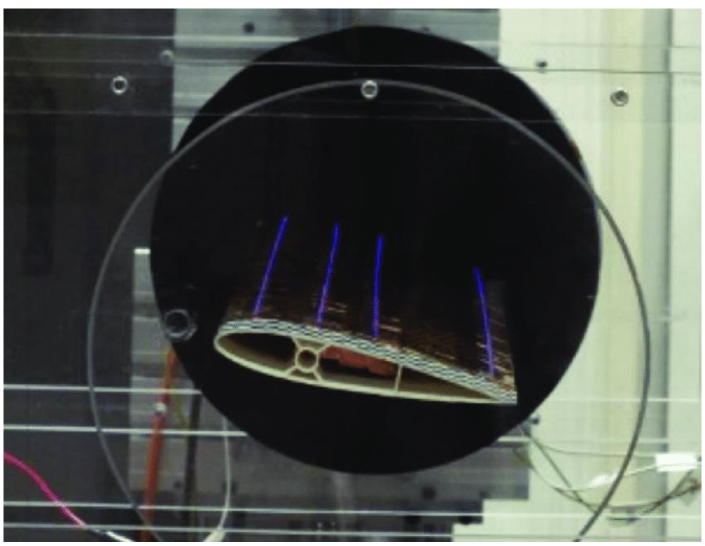


Fig 6.5. *Airfoil equipped with four spanwise plasma actuators* [14]

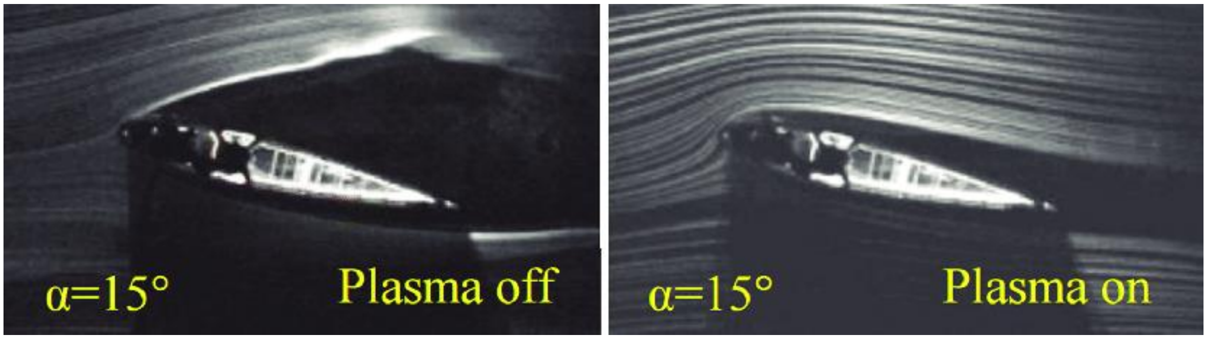


Fig 6.6. *Flow structure around NACA0015 airfoil at Re = 15000 without (left-hand side) and with (right-hand side) plasma actuation* [14]

6.3. PROS AND CONS OF A CONVENTIONAL DBD PLASMA ACTUATORS

Studies carried out in the last decades have demonstrated the possibility to use DBD aerodynamic actuators for active flow control purposes. The main advantages and disadvantages of the use of these plasma devices are summarized below:

Pros

• Possibility to locate these actuators in different positions on a surface and over existing aerodynamic bodies.
• Negligible aerodynamic interferences when they are not active.
• Fast actuation times.
• Low power consumption.
• Possibility to tune the on/off actuation strategy depending on the particular fluid dynamic condition.

Cons

• Induced velocity below 10 m/s.

• Low electric into kinetic conversion efficiency.

• Use of potentially hazardous high voltages and electromagnetic noise generation.

• Effectiveness in flow-control for Reynolds numbers typically below a few hundred thousand.

Disadvantages already introduced will be overcome only when more detailed knowledge of basic interactions between discharge a neutral gas will be achieved. When higher induced speeds will be available, new and more effective flow control strategies will be developed.

*****

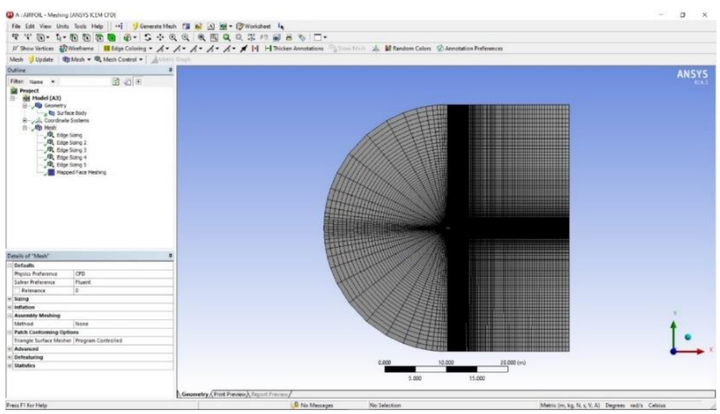
CHAPTER 7

MODELLING AND ANALYSIS OF A DBD PLASMA ACTUATOR

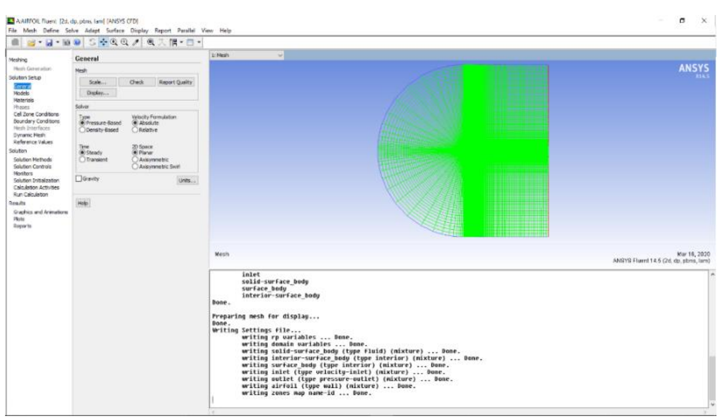
7.1. ANALYSIS OF PRESSURE COEFFICIENT OF NACA 23015 AIRFOIL IN STEADY LAMINAR FLOW WITH AND WITHOUT DBD PLASMA ACTUATORS

The following analysis was done in ANSYS Workbench and a C type mesh was generated for the purpose of analysis as this is seen to be more accurate for the analysis of airfoils as shown in [Figure 7.1].

A C-type mesh pattern captures the leading-edge curvature without any singularities. Though the C-type pattern avoids the propagation of boundary layer fineness upstream, it fails to do so in the downstream of the airfoil trailing edge. In a way, this downstream fineness proves beneficial as it helps to capture the shear layer for solver runs at low angles of attack.



(a)



(b)

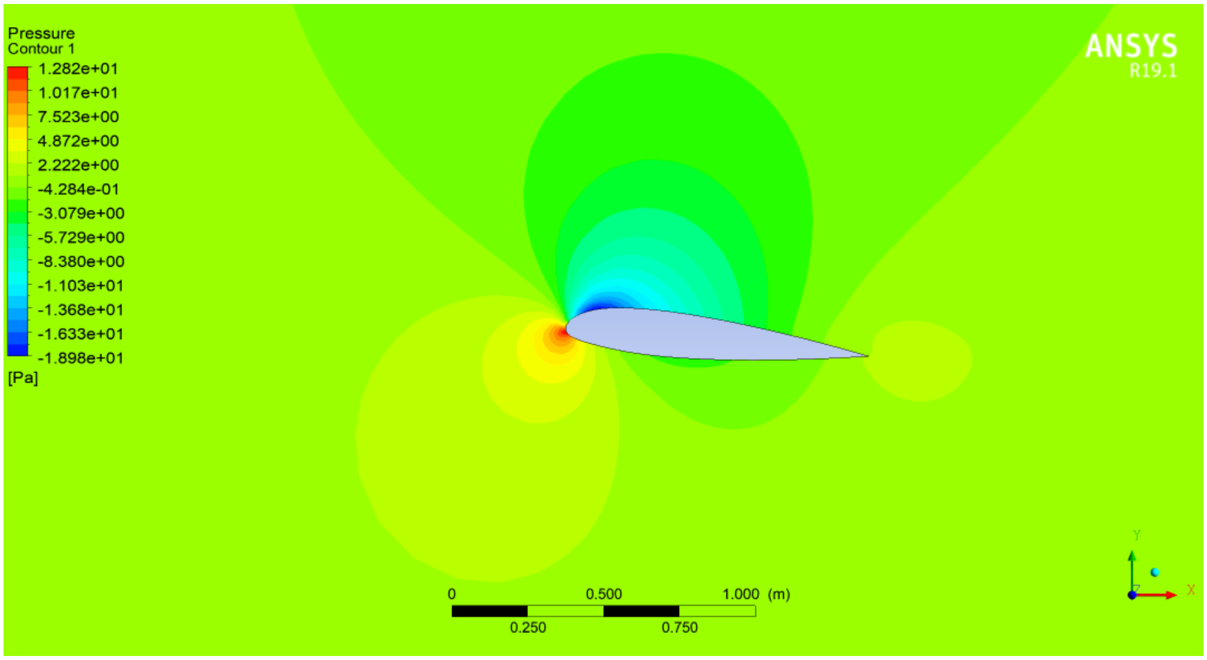
Fig 7.1. (a) and **(b)** *C-Type mesh generation of an airfoil with 5° angle of attack*

In the figures given below, the pressure coefficient contours are obtained over NACA 23015 airfoil with a 5° angle of attack. In [Figure 7.1(a)], the plasma actuator is switched off, and the flow separation begins to occur. When the plasma actuator is activated, reattachment of the flow is achieved as shown in [Figure 7.2(b)].

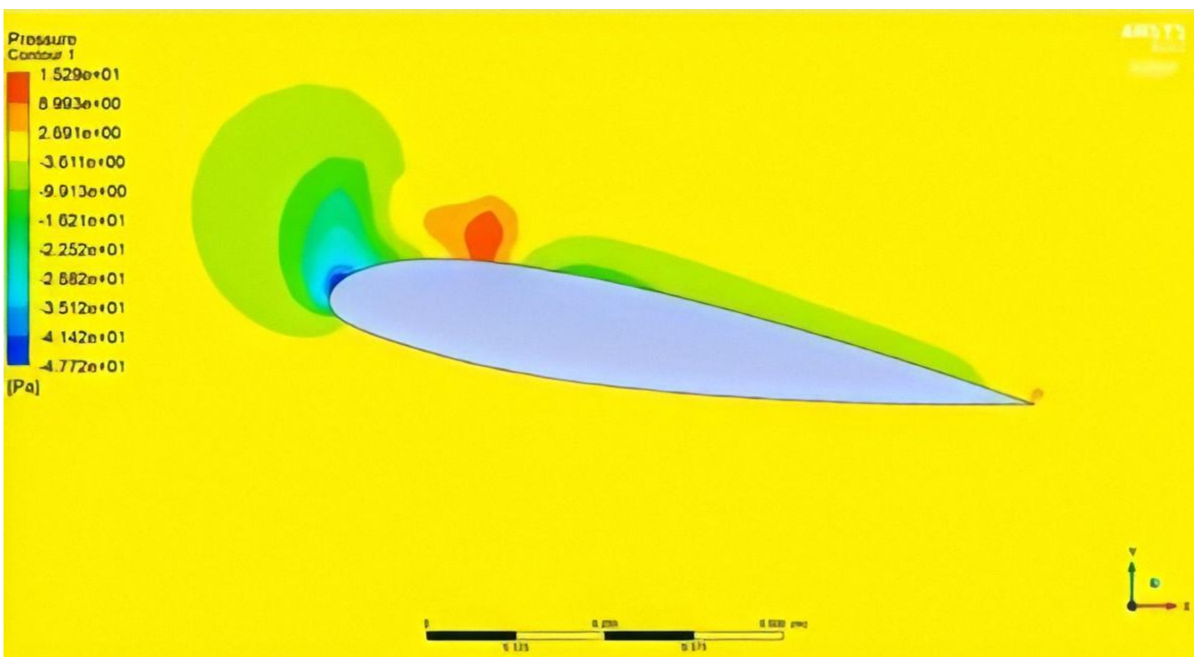
The NACA23015 airfoil is equipped with spanwise vectorized plasma actuators located at different airfoil positions.[63] When actuators are continuously operated, lift increments are limited to about 15%. When operated with a duty cycle strategy, lift increments are close to 50%.

Plasma actuators have demonstrated to strongly improve their ability in flow manipulation when operated with a duty cycle strategy. With this approach, the DBD device is turned on and off with intermittence by following a particular duty cycle frequency. This frequency is usually chosen in the range 5–100 Hz and it is strictly related to natural vorticities developing over the aerodynamic surface. The percentage ratio between the period in which discharge is fed and the whole duty cycle period is called duty cycle percentage. If it is fixed to 50%, it means that discharge is ignited half of the time.

On a parallel plane, this actuation strategy leads to lower power consumption. Plasma actuators can also be used to reduce noise, induced by the aerodynamic surfaces, especially by landing gears and trailing edges [64–66]. Studies on bluff bodies have demonstrated the ability of DBD actuators to reduce downstream turbulence, leading to the suppression of particular tones or overall noise mitigation up to 4dB.



(a)



(b)

Fig 7.2. *Pressure coefficient contours of NACA 23015 airfoil with 5° angle of attack: plasma actuator off (a) and on (b)*

7.2. MODELLING OF A DBD PLASMA ACTUATOR ON A LANDING GEAR STRUT USING CATIA V5

The plasma actuator consists of an exposed electrode constructed from 74µm thick tinned copper foil which is 5 mm wide. The encapsulated electrode is constructed from the same material but 10 mm wide. Kapton tape is layered on top of the encapsulated electrode which acts as the dielectric between the electrodes. Kapton is a polyimide film developed by DuPont in the late 1960s that remains stable across a wide range of temperatures, from −269 to +400 °C.

Our Nose Landing Gear which was previously designed is now incorporated with the DBD Plasma Actuator and is modelled using the same modelling software previously used i.e. CATIA V5. The steps involved in designing the same is given below:

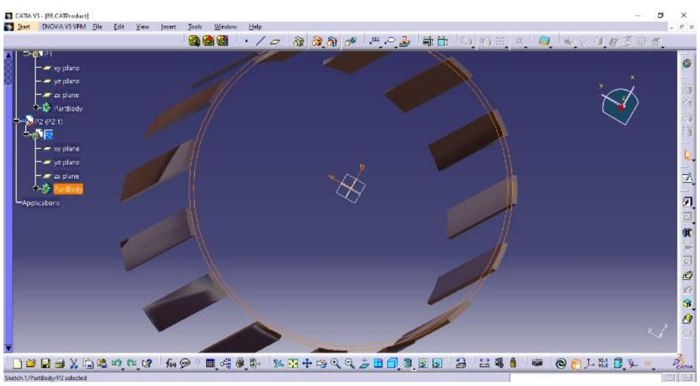


Fig 7.3.*Padding the exposed electrode profile and patterning (Top view)*

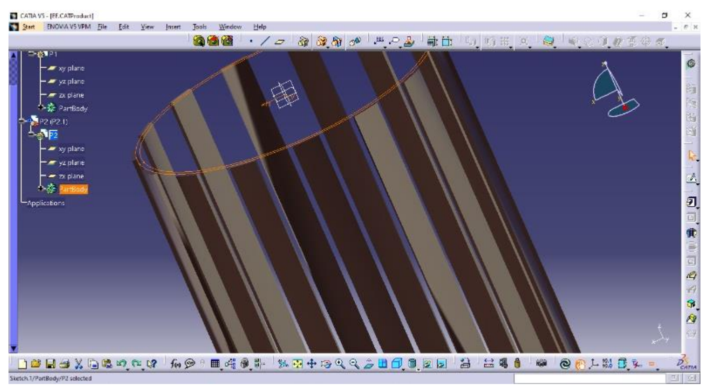


Fig7.4. *Exposed electrodes (Isometric view)*

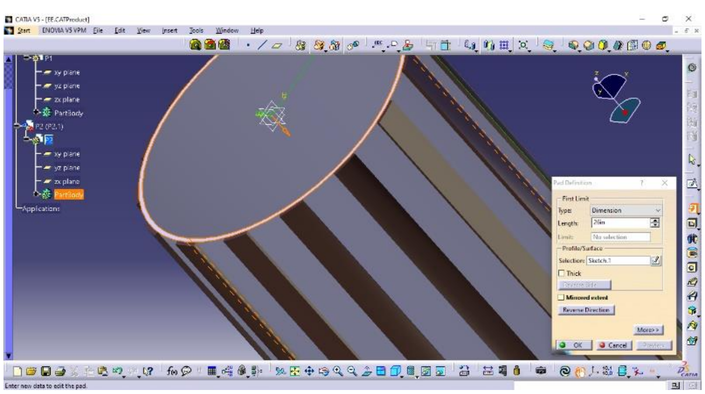


Fig 7.5. *Padding the inner dielectric material (Isometric view)*

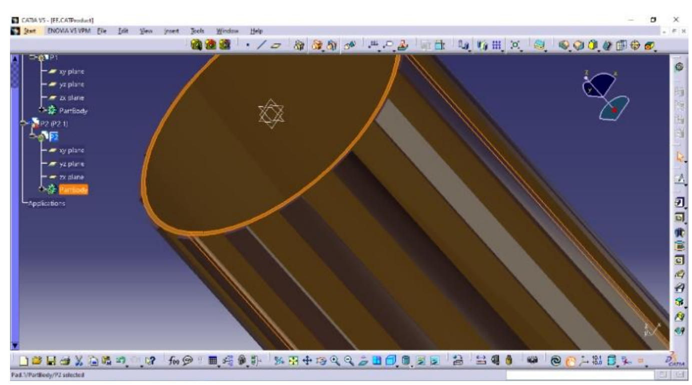


Fig 7.6. *Exposed electrodes along with dielectric profile (Isometric view)*

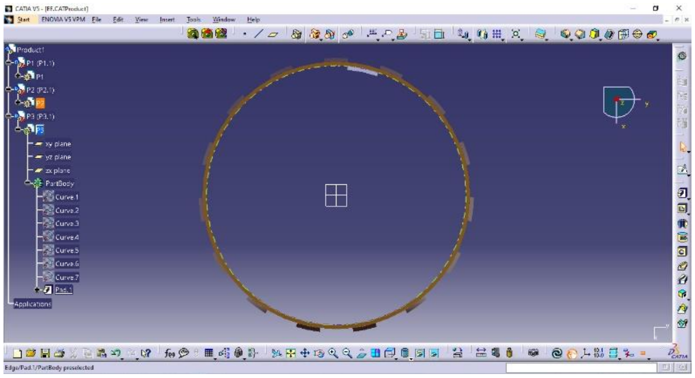


Fig 7.7. *Padding the covered electrode profile and patterning (Top view)*

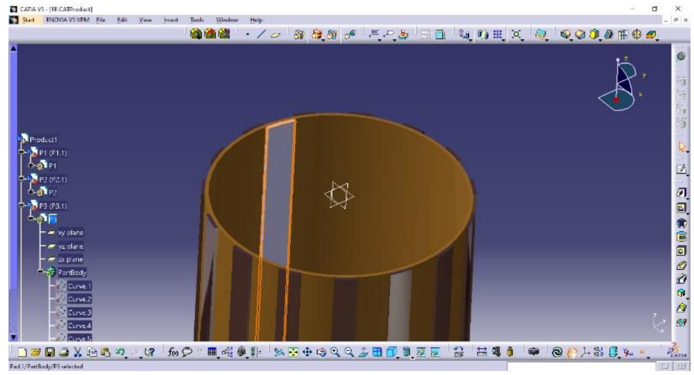


Fig7.8.*Covered electrode(Isometric view)*

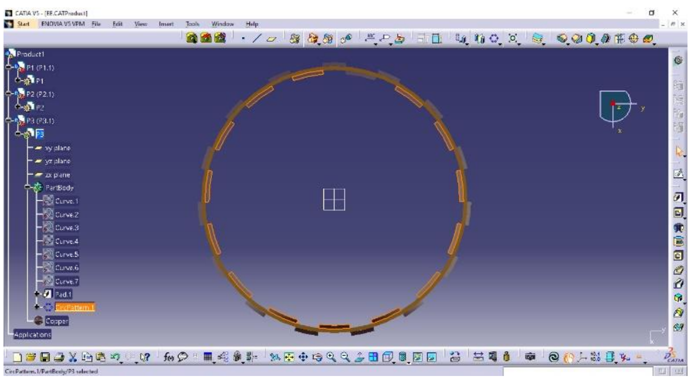


Fig 7.9. *Circular pattern of covered electrode (Top view)*

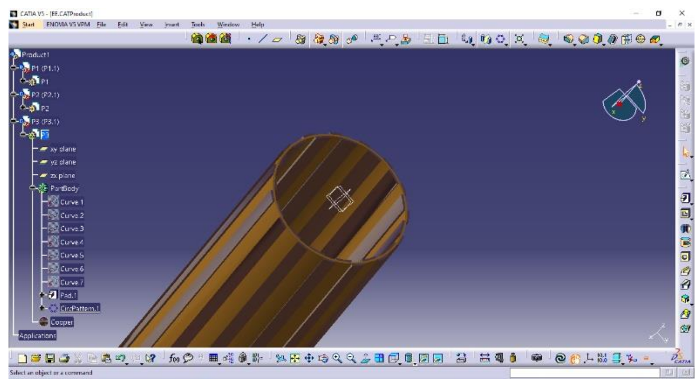


Fig 7.10. *Combination of electrodes and dielectric (DBD Plasma Actuator)*

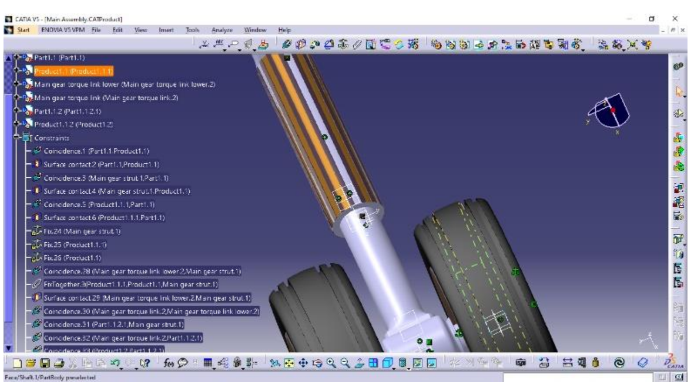


Fig 7.11. *Assembly of DBD Plasma Actuator on landing gear upper strut*

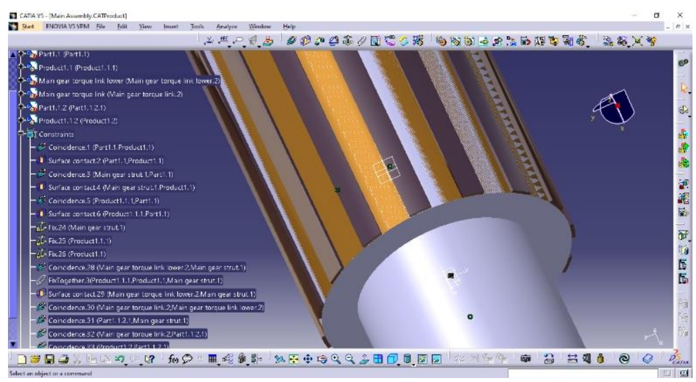


Fig 7.12. *Assembly of DBD Plasma Actuator on landing gear upper strut (Close up view)*

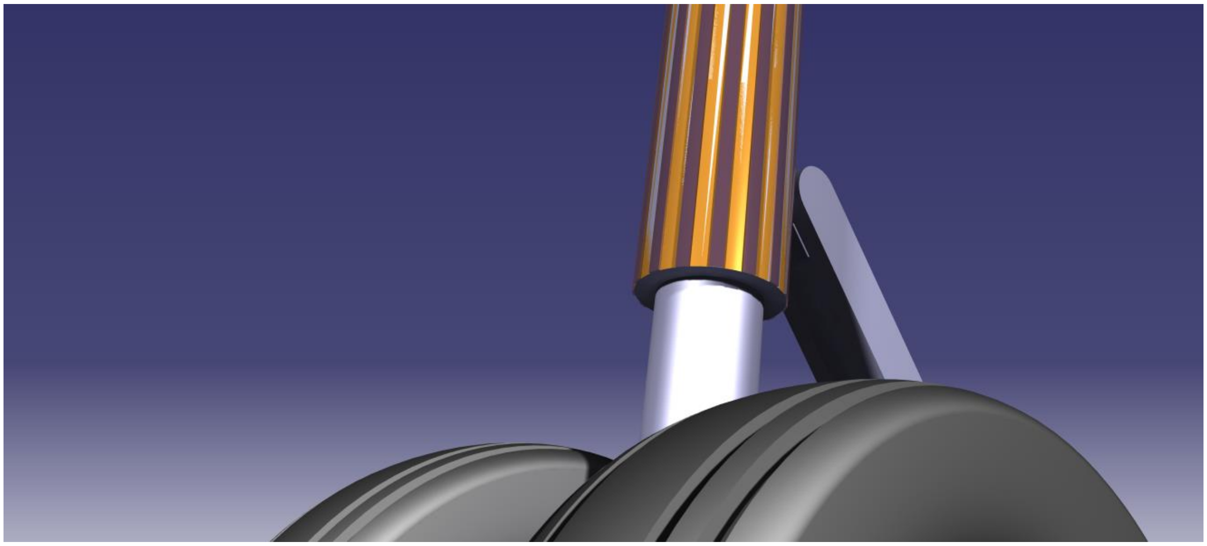


Fig 7.13. *Final Rendered view of the Nose Landing Gear fitted with the multi-electrode DBD Plasma Actuators (Close up view)*

Fig 7.14. *Final Rendered view of the Nose Landing Gear fitted with the multi-electrode DBD Plasma Actuators (Isometric view)*

7.3. MODELLING OF LANDING GEAR STRUT WITH DBD A PLASMA ACTUATOR FOR THE PURPOSE OF ANALYSIS

Many of the components that form the landing gear take the form of bluff bodies in cross-flow. In this study proof-of-concept, experiments are presented which serve to demonstrate the ability of plasma actuators to reduce bluff body flow separation that is a source of landing gear noise.

For the simplicity of analysis, the nose landing gear strut is considered as a 2-D cylinder when viewed from the top view. This is then modelled with the DBD plasma actuators to reduce flow separation about the element. This effective streamlining of the landing gear element by the plasma actuators is termed a ― plasma fairing. An example is shown in [Figure 7.15(a)] where a generic landing gear strut is shown.

Both upstream and downstream components of the nose landing gear are covered with an array of plasma actuators like the one shown in the figure below, which are capable of producing locally tangential velocity. The electrodes extend along the strut. A schematic of the flow in the cross-cut plane A-A′ without the use of actuators is shown in [Figure 7.15(b)].

The flow is characterized by large-scale separation over the leading and trailing elements, the formation of unsteady large-scale vorticity in the wake that is subsequently distorted by the downstream strut element. This interaction, along with the pressure fluctuations at the surface of the upstream element due to Karman vortex shedding, gives rise to an effective dipole source for acoustic radiation. [68]

A schematic of the flow with the actuator array is shown in [Figure 7.15(c)]. As shown in the inset panel, the surface-mounted electrodes are operated to accelerate fluid along the surface, giving rise to a local wall jet effect. This wall jet tends to remain attached to a convex surface. This phenomenon is commonly referred to as the Coanda effect. [67]

In both steady and unsteady mode, the plasma actuator causes an intensive turbulent mixing in the initially laminar boundary layer. This serves to transfer momentum from high velocity to low velocity layers making the boundary layer much more stable and drastically delaying separation. [68]

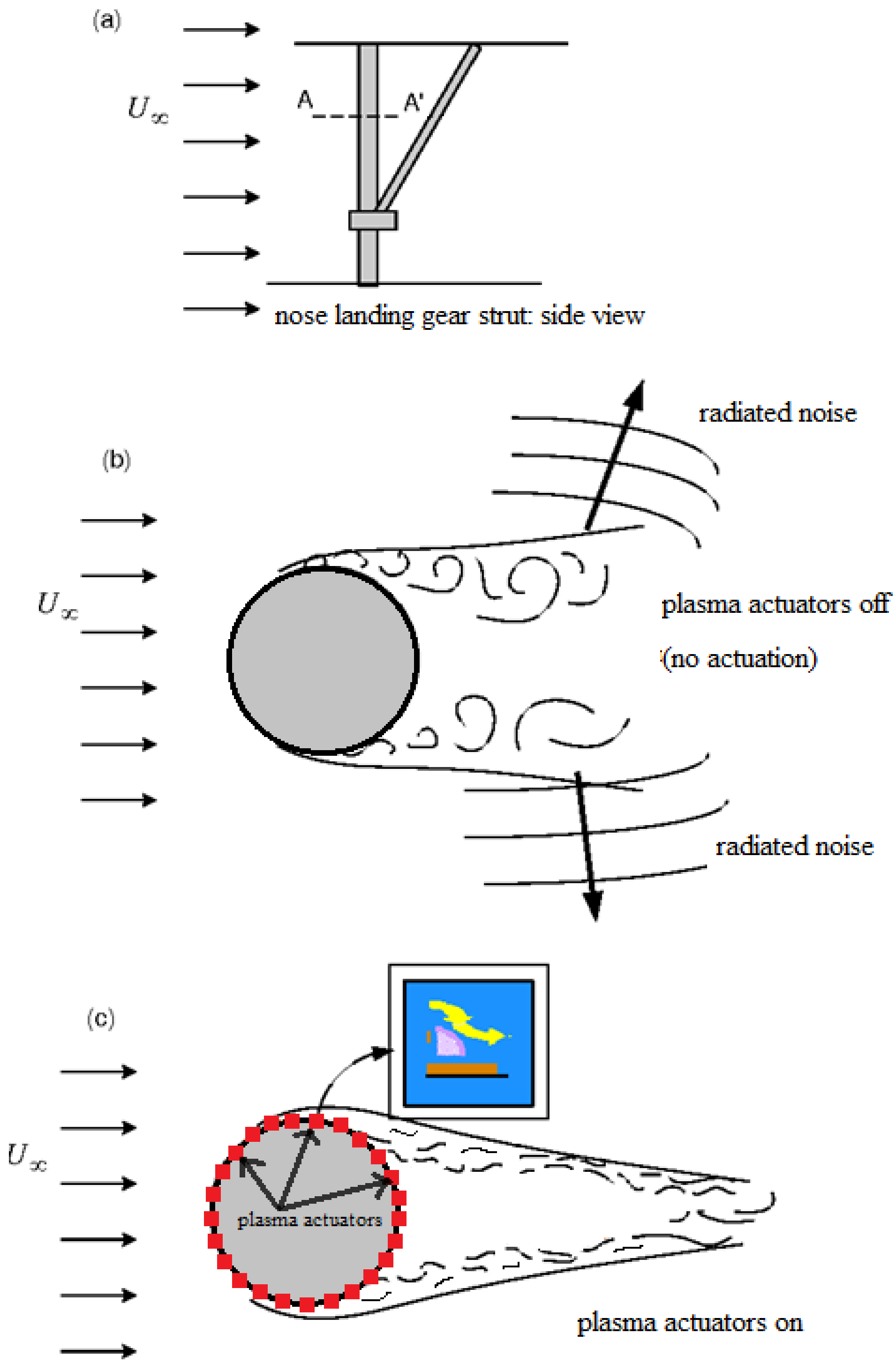
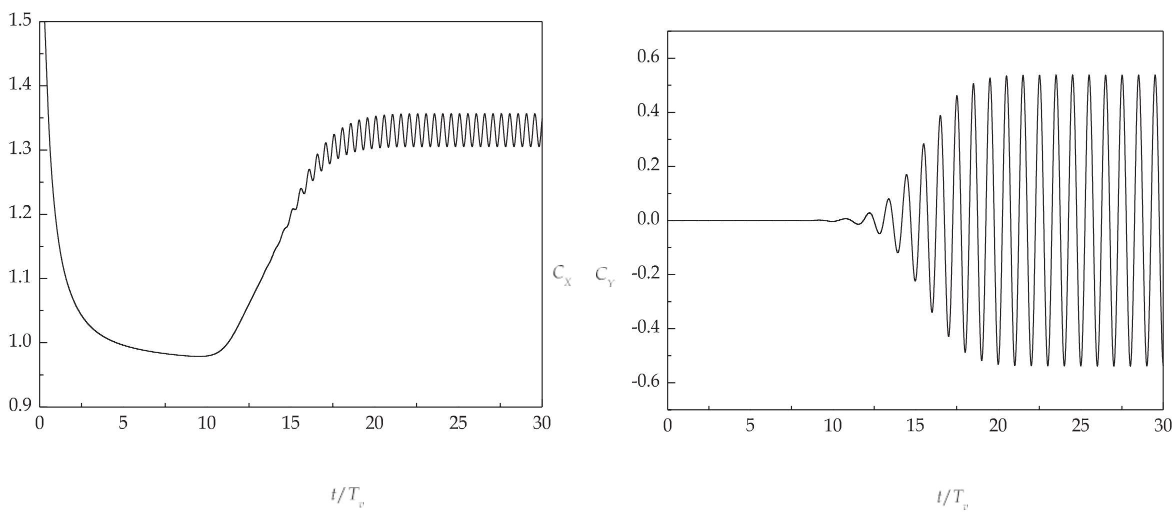


Fig 7.15. *Schematic of a plasma actuator array on a landing gear strut.*

7.4. ANALYSIS OF CIRCULAR CYLINDER IN UNSTEADY TRANSIENT FLOW SHOWING THE VON-KARMAN EFFECT

In this section, results are shown for steady incident flow. The evolution of the instantaneous drag and lift coefficients is shown in [Figure. 7.16] for a typical run at *Re* = 150. In this simulation, the velocity was set to zero everywhere in the flow domain. It takes more than ten periods of vortex shedding for the flow oscillations to be initiated in the wake. During this period, the instantaneous drag drops rapidly from very high values, then reaches a plateau before rising again in [Figure 7.16(a)]. Once the instability is triggered, the oscillations are amplified and become saturated in approximately another ten periods as it is seen more clearly in the history of the lift in [Figure 7.16(b)]. Beyond this point, a steady-state periodic oscillation is established. [69]



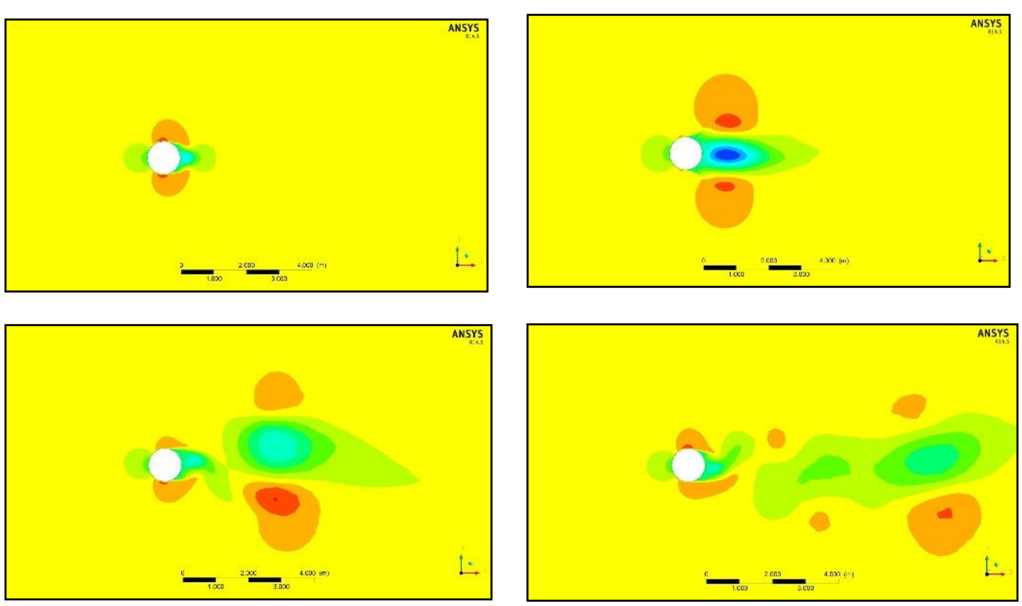
(a) Drag history (b) Lift history

Fig 7.16. *Instantaneous in-line (drag) and transverse (lift) forces for Re = 150.*

where θ is the angle measured clockwise from the front stagnation point. The force is nondimensionalized to yield the in-line (drag) and transverse (lift) force coefficients according to formulae given below:

$$C_X = \frac{F_X}{\frac{1}{2}\rho u_\infty^2 D} \qquad\qquad C_Y = \frac{F_Y}{\frac{1}{2}\rho u_\infty^2 D}$$

[Figure 7.17] shows the vorticity distribution around the cylinder at six instants over approximately half vortex shedding cycle in the steady periodic state. The instantaneous forces on the cylinder are also shown at the bottom subplots and symbols (red circles) indicate the different instants shown. The figure shows the processes of vortex formation and shedding in the near wake. Vortices are regions of low-pressure and as they are formed near the cylinder, a fluctuating pressure field acts on the cylinder surface. As a result, the drag is minimized while a positive vortex (marked by red contour lines) is formed on the bottom side of the cylinder in phases 1 and 2. Subsequently, this positive vortex moves towards the opposite side and away from the cylinder and the drag increases to its maximum in phase 4. In phases 5 and 6, a negative vortex begins to form on the top side of the cylinder which, in turn, will induce a low back pressure and a minimum in drag as it gains strength and size. A maximum in the lift is observed between phases 3 and 4 which is associated with the entrainment of the fluid at the end of the formation region across the wake centerline and towards the top side which causes the negative vortex on the same side to be shed. This process repeats periodically and a staggered vortex street is formed downstream in the wake. [69]



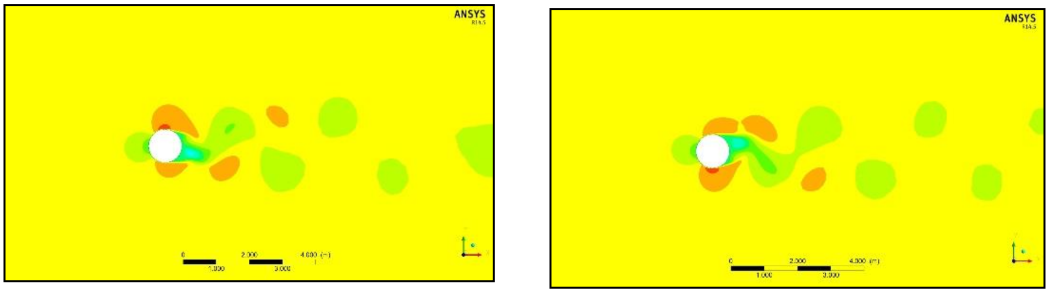
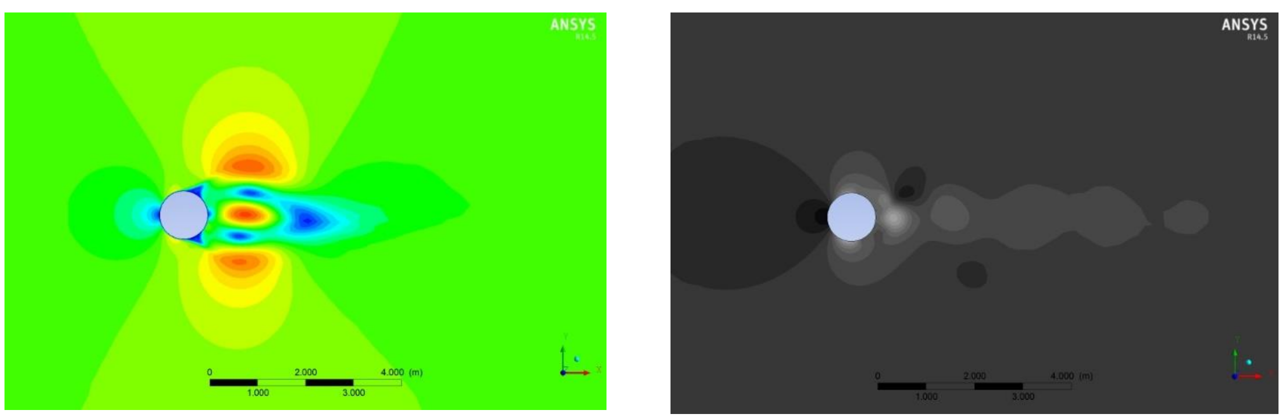


Fig 7.17. *Vorticity distributions in the cylinder wake and fluid forces on the cylinder at different instants in unsteady transient flow; Re = 150.*

Velocity is normalized with the incident flow velocity, u_∞, and pressure is normalized with $\frac{1}{2}\rho u_\infty^2$. The distribution of the mean streamwise velocity shows the acceleration of the fluid as it passes over the 'shoulders' of the cylinder and the 'recirculation bubble' (reverse velocities) behind it [Figure 7.18(a)]. The bubble closure point where the mean velocity becomes zero is located 1.6 diameters behind the cylinder centre.[69] When Bernoulli's equation is applied to the flow outside separation, the back-pressure coefficient is given by (Griffin & Hall, 1991) as:

$$C_{pb} = \frac{p_b - p_\infty}{\frac{1}{2}\rho u_\infty^2} = 1 - K^2$$



(a) *Streamwise mean velocity* ***(b)*** *Mean pressure*

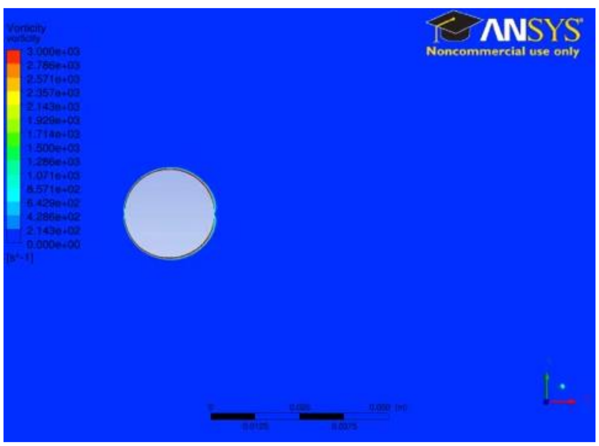
Figure 7.18. *Time-averaged statistics of the velocity and pressure fields at Re = 150.*

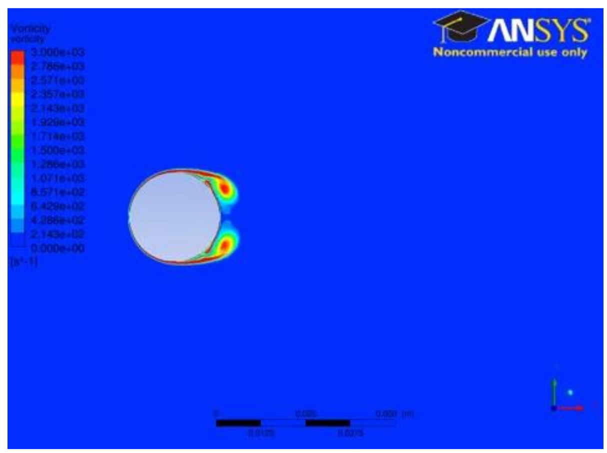
7.5. ANALYSIS OF CIRCULAR CYLINDER WITH DBD PLASMA ACTUATOR IN UNSTEADY TRANSIENT FLOW USING ANSYS WORKBENCH

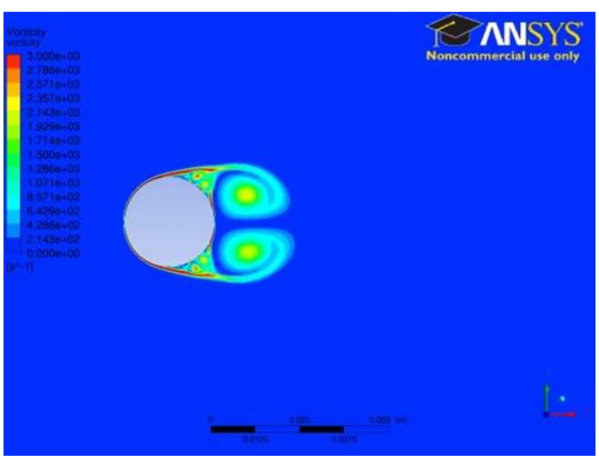
The following analysis was done on a 2D circular cylinder fitted with the DBD Plasma Actuator along the circumference of the circle using ANSYS Workbench.

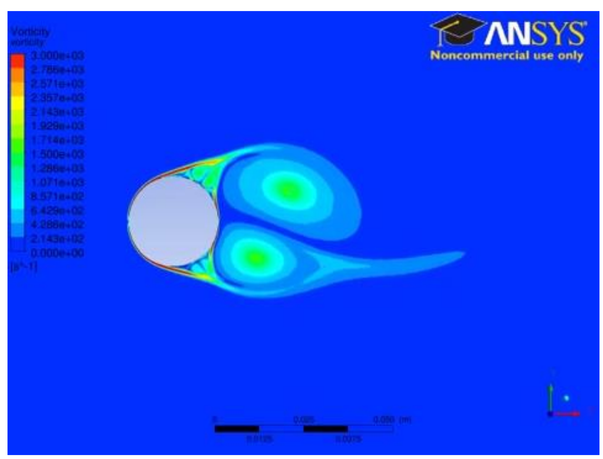
The results obtained using the analysis clearly shows the disappearance of the Von-Karman Effect which is observed on the cylindrical bluff body when the Plasma Actuators are turned on as compared to in its off position.

7.5.1 ANALYSIS WHEN PLASMA ACTUATORS IS TURNED OFF









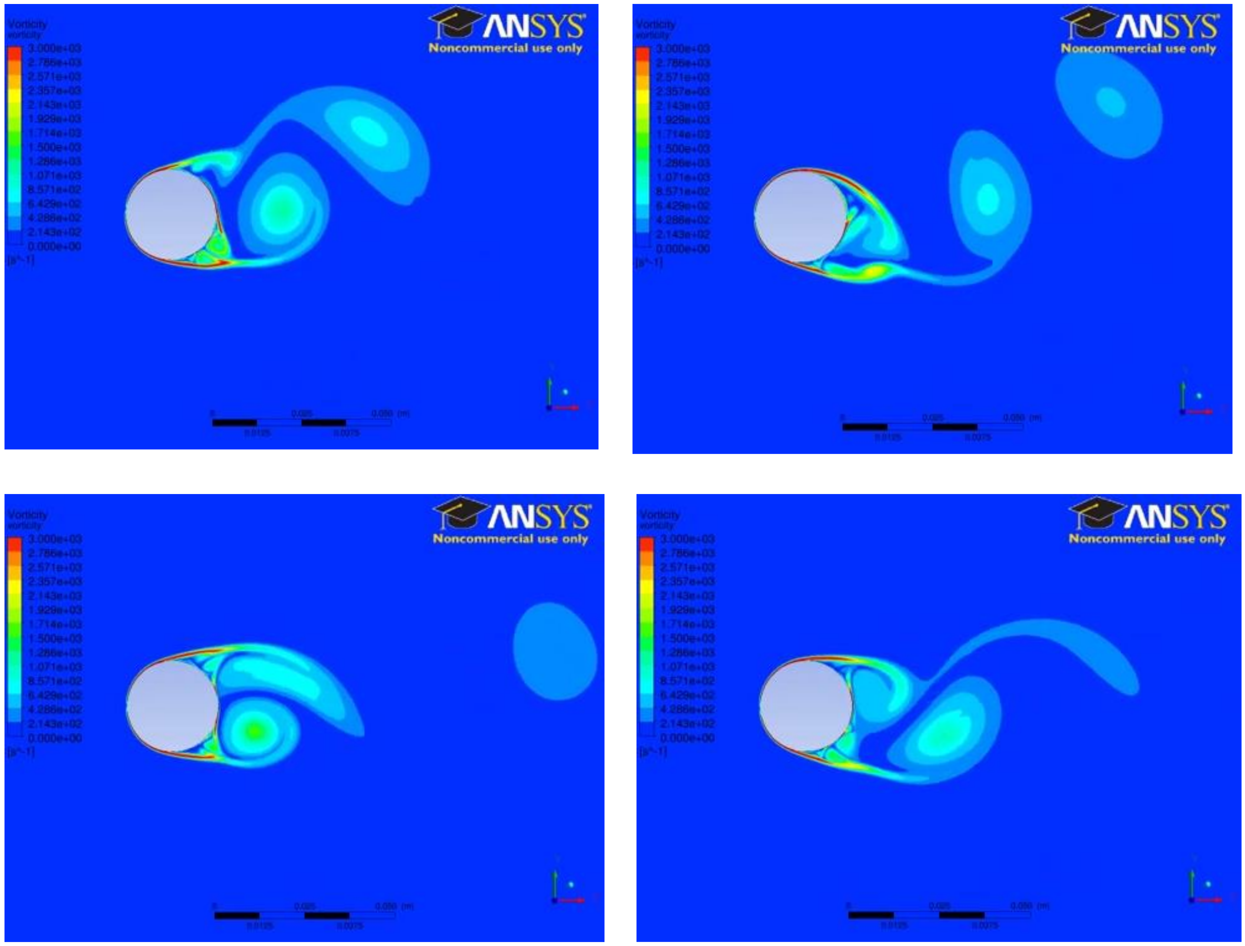
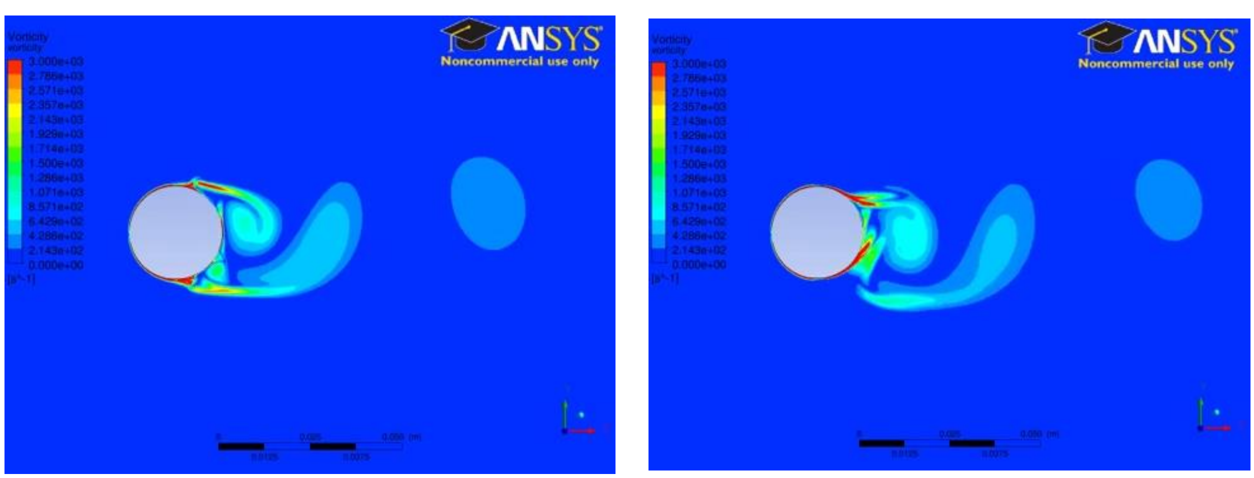


Fig 7.19. *Transient flow analysis of 2D cylinder with DBD Plasma Actuator* ***off*** *showing the presence of Von-Karman effect with a step size of 0.5sec.*

7.5.2. ANALYSIS WHEN PLASMA ACTUATORS TURNED ON



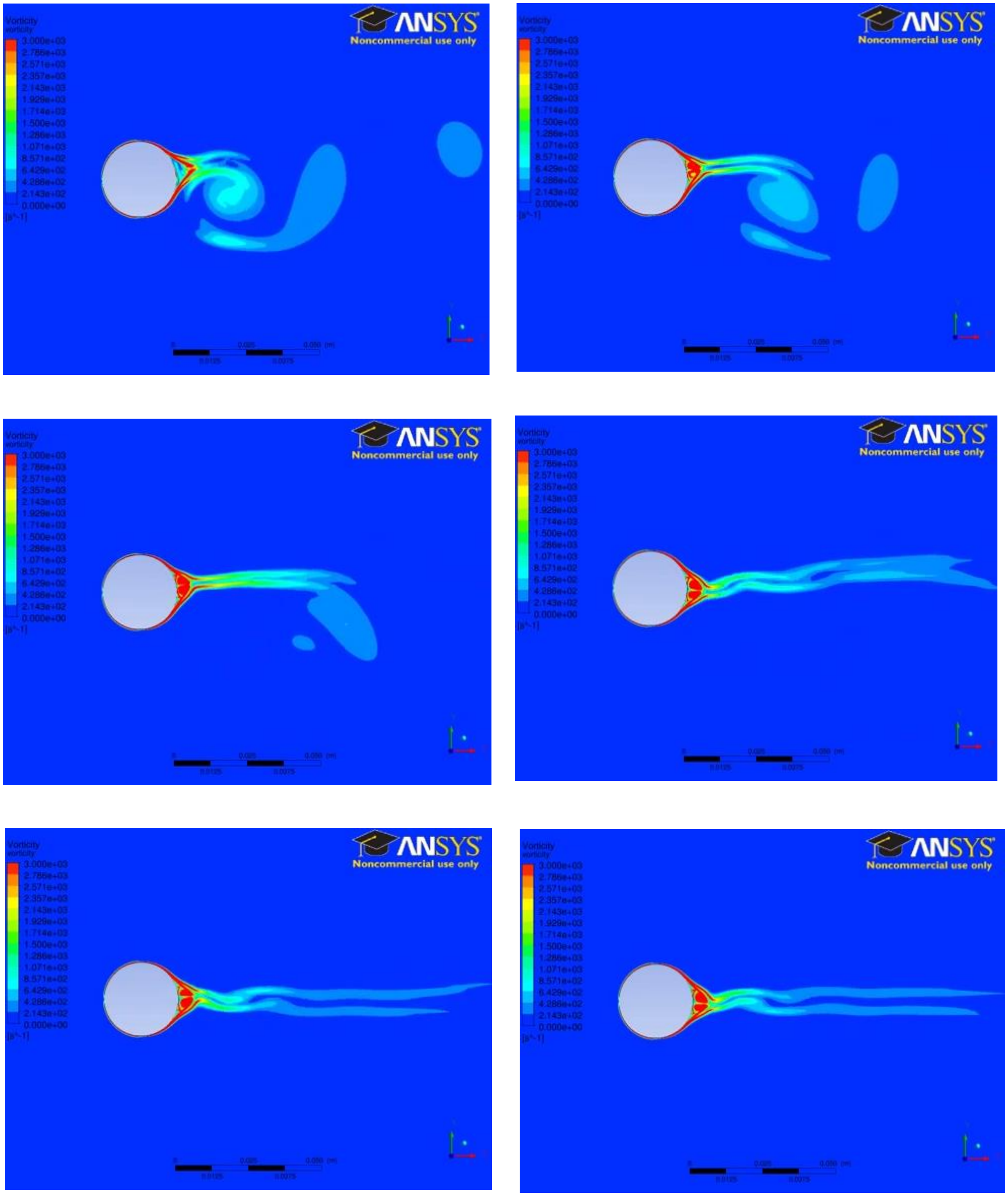


Fig 7.20. *Transient flow analysis of 2D cylinder with DBD Plasma Actuator* ***on*** *showing the absence of Von-Karman effect with a step size of 0.5sec.*

7.6. POWER CONSUMPTION OF DBD PLASMA ACTUATORS

In a study published by Rodrigues, Frederico & Páscoa, José & Dias, Filipe in their research paper “Power Consumption Characterization of DBD Plasma Actuators for boundary layer control.” the power consumption of DBD plasma actuators was estimated for actuators with different dielectric materials: 0.3 mm thickness of Kapton, 0.6 mm thickness of Kapton and 1.12 mm thickness of polycarbonate and Kapton (1mm of polycarbonate plus 0,12 mm of Kapton). Using the voltage and current data, the power consumption was computed.

They found that the actuator made up of 0.3 mm of Kapton consumed more power than the other ones. In the first attempt a plasma actuator made up of 1 mm polycarbonate layer was used, however when the plasma formation started the polycarbonate began to melt. After that, a plasma actuator with a dielectric layer composed of two layers of Kapton above 1 mm of polycarbonate layer was used. They verified that this dielectric layer allows the operation of the plasma actuator

at high voltage levels without the polycarbonate layer melting, supporting higher levels of voltage than the actuator of 0.3 mm Kapton. From the actuators tested in this work, the actuator made of

polycarbonate and Kapton was the actuator with lower power consumption. In this actuator, due to the total thickness of the dielectric, the plasma formation only happens for an applied voltage above to 5 kV.

7.7. INFLUENCE OF FREQUENCY AND VOLTAGE WAVEFORM

Initially, the thrust grows very fast (Figure 7.21), with voltage magnitude increasing, but at a certain point, it stops growing, although the power dissipation (Figure 7.22) keeps going up. This body force saturation is accompanied by a visible change in the discharge structure. In addition to the blue uniform glow, bright filaments appear, as shown in (Figure 7.23). At high frequency (8 kHz), the saturated discharge contains many small filaments, while at the low frequency (1 kHz) the discharge has a small amount of very large filaments. Further increases in voltage, beyond the saturation point, result in an increase of the filament intensity, very high heat dissipation, dielectric degradation, and, finally, the plasma actuator 's breakdown. The dotted line in Figure 7.21 connects all the saturation points for the sinusoidal voltage. The dependence of the thrust on the saturation voltage at the varying frequency is close to linear. [79]

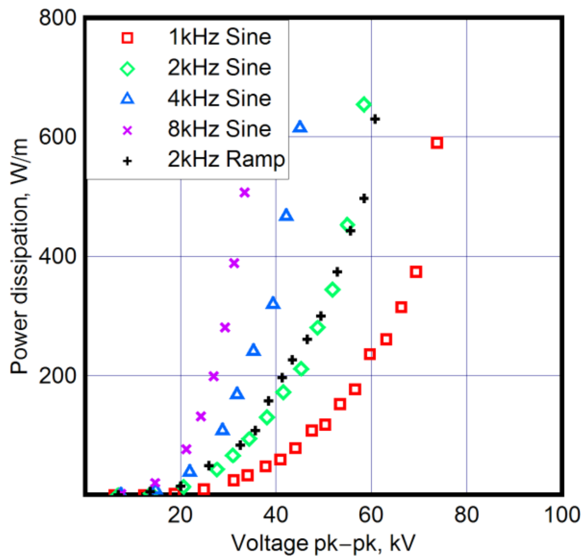


Fig 7.21. *Effect of frequency and voltage waveform on power dissipation. (0.125in. quartz glass).*

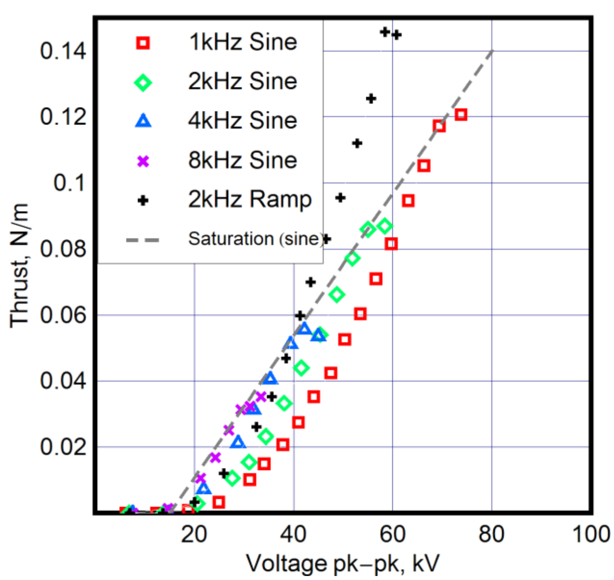
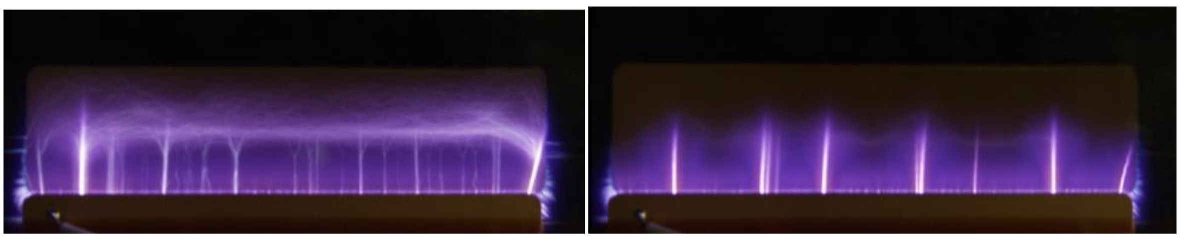


Fig 7.22. *Effect of frequency and voltage waveform body force (0.125in. quartz glass).*

Figure 7.20 shows the dependences of the power dissipation in the plasma actuator on the peak to peak voltage magnitude for the same plasma actuator at different frequencies and voltage waveforms. The dielectric is quartz with a 1/8-inch thickness. The power dissipation at a given voltage is roughly proportional to the frequency. [79]

The dependence of thrust on the voltage magnitude for a 1/8-inch-thick quartz actuator is shown in Figure 7.21. Higher frequencies produce higher body force, similar to the power dissipation behaviour in Figure 7.20.



(a) **(b)**

(c) **(d)**

(e)

Fig 7.23. *Filamentary structure of plasma discharge at various frequencies for 0.125in. quartz glass: (a) 1 kHz, (b) 2 kHz, (c) 4 kHz, (d) 8 kHz, (e) normal discharge at 1 kHz.*

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CHAPTER 8

PROPOSED SYSTEM

8.1. ENHANCED MULTI-ELECTRODE DBD PLASMA ACTUATOR

The concept of our proposed system is based on Electrohydrodynamic force. (EHD) force is used for active control of fluid motion and for the generation of propulsive thrust by inducing ionic wind with no moving parts. The proposed method successively generates and accelerates ionic wind induced by surface dielectric-barrier-discharge (DBD), referred to as a DBD plasma actuator with multiple electrodes. A conventional method fails to generate unidirectional ionic wind, due to the generation of a counter ionic-wind with the multiple electrodes DBD plasma actuator. However, unidirectional ionic wind can be obtained by designing an applied voltage waveform and electrode arrangement suitable for the unidirectional EHD force generation. Our results demonstrate that mutually enhanced EHD force can be generated by using the multiple electrodes without generating counter ionic-wind and highlights the importance of controlling the dielectric surface charge to generate the strong ionic wind. The proposed method can induce strong ionic wind without a high-voltage power supply, which is typically expensive and heavy and is suitable for equipping the aircraft and small unmanned aerial vehicles (UAV's) with a DBD plasma actuator for a drastic improvement in the aerodynamic performance.[70]

In order to improve the performance of DBD plasma actuators, optimization studies in view of electrical and fluid properties have been conducted [71]. A straightforward method of enhancing the induced flow velocity is to increase the applied voltage amplitude. Some previous studies show that the induced flow velocity increases monotonically with increasing voltage but then becomes saturated at high voltage levels [72,73]. The maximum induced velocity reaches 6 m/s when the peak-to-peak value of the applied voltage is 50 kV [74]. A critical drawback to increasing the amplitude of the applied voltage is that a high-voltage power supply is needed, and it is usually expensive and heavy on aircraft. However, equipping aircraft with high-voltage power supplies hinders their practical applications. In addition, considerable isolation distance between the DBD plasma actuator and the other electronic devices is required to avoid unexpected discharge when high voltage is applied to the DBD plasma actuator. This requirement is also a drawback for practical applications.

Another method of enhancing EHD force involves using several DBD plasma actuators in series. A previous experimental study observed the additional acceleration of ionic wind and a maximum velocity of approximately 7 m/s, measured using four actuator modules (a single actuator module is composed of two electrodes, i.e., a single DBD plasma actuator) with a voltage peak-to-peak value of 40 kV and a frequency of 1 kHz [74]. However, there must be a gap between each actuator module, because counter ionic-wind is induced when successive actuator modules are close together or when high voltage is applied, degrading the performance of the integrated DBD plasma actuator. This is known as the cross-talk phenomenon [76]. The need for a gap restricts the integrated density of the actuator modules and impedes efficient acceleration of ionic wind. Counter ionic-wind is generated as a result of the discharge between a covered electrode and the exposed electrode of adjacent modules as shown in Fig. 1b. This counter ionic wind collides with the main flow and deflects wind in the vertical direction to the dielectric surface when there is no gap between each actuator module, as in the case of plasma synthetic jet actuators [77]. The generation of the counter ionic-wind due to the cross-talk phenomenon is undesirable given the multiplied acceleration of ionic wind by successive actuator modules.

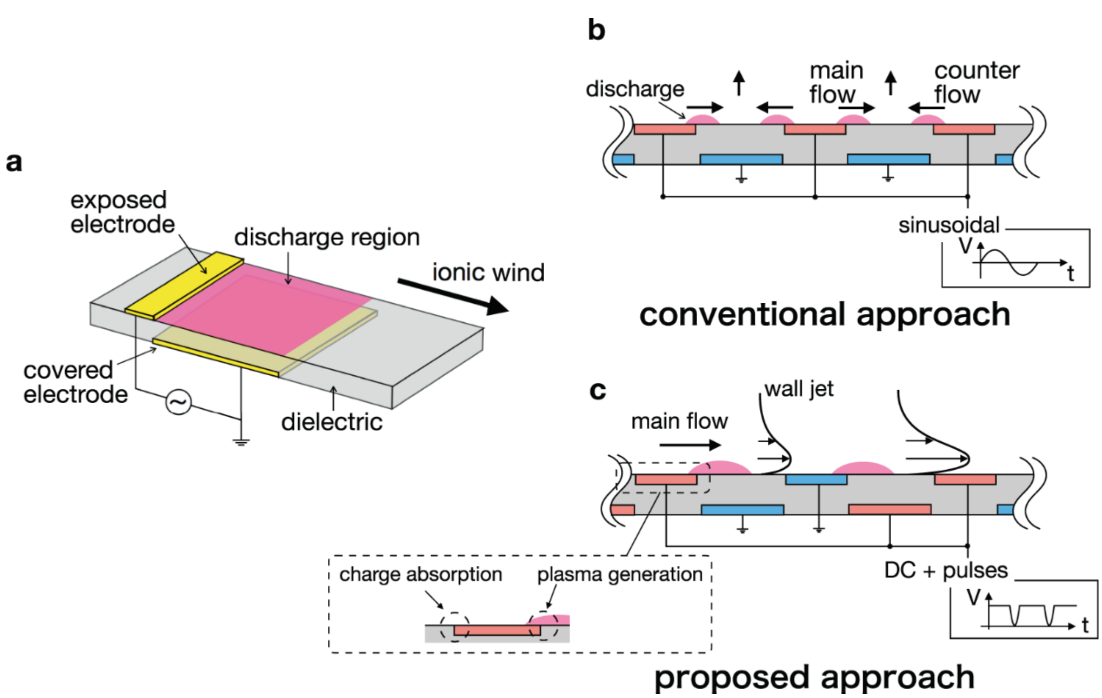

Fig 8.1. *Schematic views of a DBD plasma actuator and method of integrating the actuator modules. (**a**) A single DBD plasma actuator composed of an exposed electrode, covered electrode, and a dielectric. Discharge occurs above the covered electrode when high voltage is applied to electrodes, generating ionic wind. (**b**) The conventional approach to integrating the actuator modules fails to monotonically accelerate the ionic wind, due to the generation of counter ionic-wind. (**c**) The proposed approach accelerates ionic wind by designing the electrode arrangement and the applied voltage such that mutually enhanced (rather than mutually degraded) EHD force is generated. In this arrangement, the downstream and upstream sides of the exposed electrode edge generate plasma and absorb the dielectric surface charge, respectively.*

In this study, we propose a multi-electrode DBD plasma actuator that mutually enhances the EHD force between successive actuator modules, rather than generating counter ionic-wind. The proposed actuator couples a particular applied voltage waveform with an electrode arrangement that is suitable for generating unidirectional EHD force. [Figure 8.1(c)] shows the proposed electrode arrangement and the applied voltage waveform.

Our approach focuses on the charge on the dielectric surface and on controlling the surface charge cycle, insofar as the surface charge plays a dominant role in the Dielectric Barrier Discharge [78]. The applied voltage waveform is DC voltage combined with repetitive nanosecond pulses rather than sinusoidal voltage.

On one hand, DC voltage is desirable for accelerating the charged particles, although the electric field generated by constant voltage is likely to be affected by electric field screening due to the charge on the dielectric surface in the DBD. On the other hand, the nanosecond pulse, which has a large voltage-slope, is suitable for generating charged particles to neutralize the surface charge. This waveform is apt for generating and accelerating charged particles. [80]

In the DBD plasma actuator driven by DC voltage combined with repetitive pulses, the charge cycle of the dielectric surface consists of two strokes: positive charging by the DC voltage, and a neutralizing positive charge by the nanosecond pulses. [78] The electrode arrangement in the proposed multi-electrode DBD plasma actuator differs from conventional approaches as follows.

In a conventional multi-electrode DBD plasma actuator, all of the exposed electrodes are connected to a power supply, and all of the covered electrodes are grounded. In our arrangement, the exposed electrodes are connected to the power supply and ground alternately, and the covered electrodes are arranged as shown in [Figure 8.1(c)] With this arrangement, counter ionic-wind is prevented because there is no potential difference between the covered electrode and the exposed electrode of the adjacent module. Consequently, no discharge is generated. Moreover, the upstream side of the adjacent exposed electrode absorbs the surface charge and prevent electric field screening, enhancing the EHD force rather than degrading it.

The potential benefit of our system is that an EHD force comparable to that of high-voltage-operation single-module actuators can be generated with low-voltage-operation highly densely integrated actuator modules because the driving voltage can be reduced by scaling down the single module and the discharge region can be extended by increasing the number of modules. The highly integrated DBD plasma actuator can actively control the airflow over the whole surface of the body, whereas the control surface is limited to the discharge region generated by two electrodes in a conventional DBD plasma actuator.

Our strategy for improving the performance of DBD plasma actuators is based on simple dynamics of surface discharge, and it has the potential to dramatically enhance the ionic wind generation with the low-voltage operation. Thus far, no study has investigated the mutual enhancement effect in a multi-electrode DBD plasma actuator with a focus on surface charge behaviour. We performed a numerical simulation of the discharge process and experimentally confirmed the mutually enhancing effect by measuring the thrust induced by the discharge and the induced flow velocity.

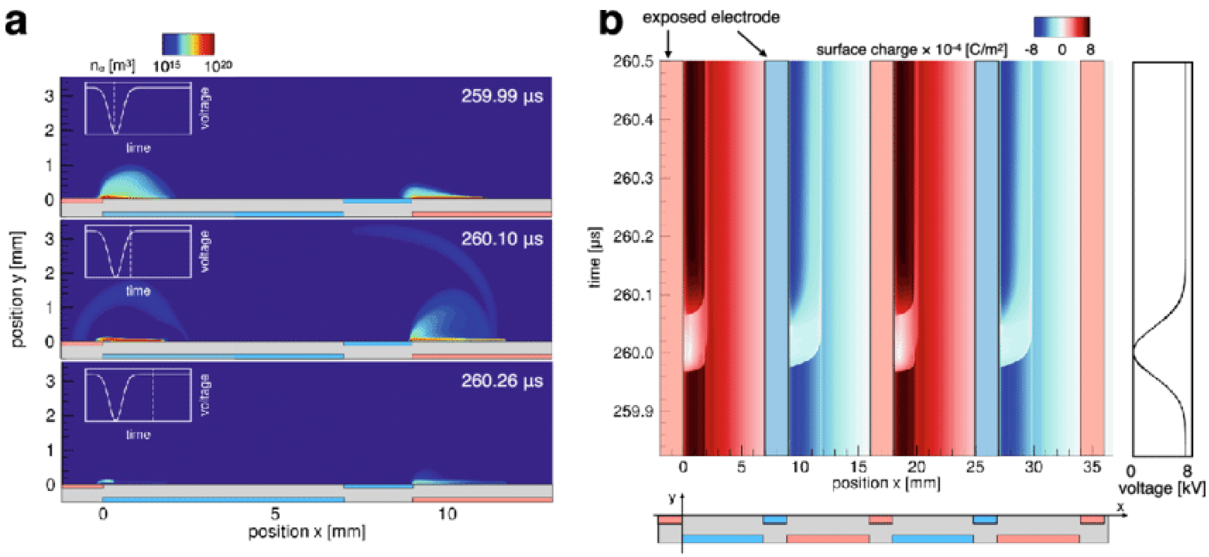

Fig 8.2. ***(a)*** *Spatial distributions of electron number density during the nanosecond-pulse phase and DC phase. In the voltage falling phase, diffusive and streamer discharges occur above the grounded and high voltage covered electrodes, respectively. By contrast, in the voltage rising phase, streamer and diffusive discharges occur above each electrode. The discharge stops after pulse superposition owing to surface charge accumulation. (**b**) Time history of the surface charge for the multi-electrode configuration. The phase of the applied voltage is shown to the right of the figure. Positive charging and neutralizing strokes form above the grounded electrode whereas negative charging and neutralizing strokes form above the high-voltage electrode around the downstream side of the exposed electrodes. In addition, the magnitude of the surface charge continuously reduces around the upstream side of the exposed electrodes.*

8.2. DISCHARGE CHARACTERISTICS

An experimental investigation of the discharge characteristics was carried out with the two- and three-electrode configurations to experimentally demonstrate the result obtained in the numerical simulation and reveal the effect of the exposed electrode downstream. DC voltage combined with repetitive nanosecond pulses was realized using a DC source and pulse generator, as shown in Figure 8.3(a). In order to prevent interference between power supplies, the DC source and the pulse generator were connected to the DBD plasma actuator via a low-pass filter and a high pass filter, respectively. [81] The power supplies were connected to the exposed electrode upstream, whereas the covered electrode and the exposed electrode downstream were connected to the ground.

A spark, which can damage the DBD plasma actuator and render the power supply unstable, occurs between the exposed electrodes in the proposed electrode configuration when the DC voltage is sufficiently high to ignite, as reported in previous studies [82,83]. In this study, a current limiting resistor (1 MΩ) was inserted between the exposed electrode downstream and the ground to stabilize the surface discharge, as shown in Figure 8.3(b). In the two-electrode configuration and the three-electrode configuration with the current limiting resistor, negative and positive current pulses were observed during the falling and rising phases of the voltage, respectively [Figure 8.3(c)]. The discharge during the falling phase occurs due to the potential difference between the exposed electrode and the dielectric surface, which is positively charged before the pulse superposition due to the DC voltage. In this phase, an electric field is generated in the direction from the dielectric surface to the exposed electrode upstream. Therefore, the negatively charged particles are spread downstream, neutralizing the positively charged dielectric surface (i.e., the neutralizing phase) [78]. In the voltage rising phase, the peak amplitude of the current pulse is smaller than that in the voltage falling phase. Contrary to the falling phase, positively charged particles spread downstream because the electric potential at the exposed electrode upstream is higher than the potential at the dielectric surface during the rising phase. These characteristics of current are similar to the current waveform when the negative nanosecond-pulse without DC voltage is applied [84], indicating that the DC component of the applied voltage has little impact on the electrical characteristics.

The peak amplitude of the current pulse during the falling phase is slightly higher in the three-electrode configuration than in the two-electrode configuration. This suggests that the electric field is enhanced due to the exposed electrode downstream. A similar trend was observed in nanosecond pulsed sliding discharge, which had a similar electrode configuration.[85] Figure 8.3(d)shows the time histories in the three-electrode configuration of the current passing through the resistor, derived from the voltage difference across the resistor using Ohm’s law. The current rapidly increases after the pulse superposition (the negative pulse is superposed at $t = 0\ \mu s$) and decays slowly, indicating that positively charged particles reach the exposed electrode downstream and transfer their charges to the ground via the resistor. This result clearly shows that the exposed electrode downstream prevents the accumulation of charge, which screens the electric field generated by the DC voltage. The peak value of the current at the exposed electrode downstream increases with an increase to the DC voltage. This is because the number of positive ions arriving at the exposed electrode downstream increases when the DC voltage increases, due to the enhancement of the electric field. It should be noted that the decay time of the current is sufficiently short to absorb the positive ions until the successive pulse superposition under the condition of this study (the frequency of the repetitive pulse is up to 2000 Hz).

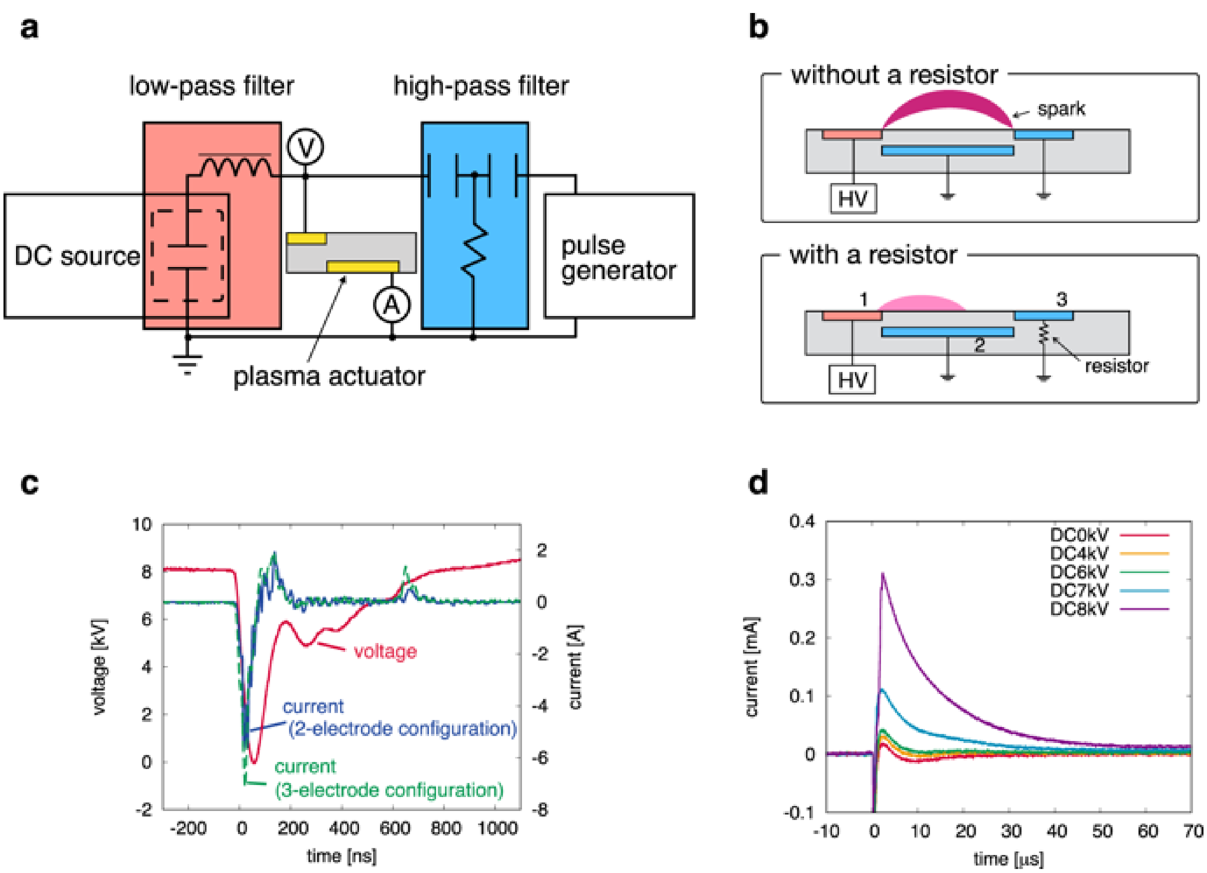

Fig 8.3. (***a***) *Schematic view of the experimental setup. The DC source is connected to the DBD plasma actuator via a low-pass filter, while the pulse generator is connected via a high-pass filter to prevent interference between power supplies.* (***b***) *Conceptual diagram of the three-electrode DBD plasma actuator with a current limiting resistor. The resistor (1*

*MΩ) is added to the exposed electrode, which plays a role in surface-charge absorption, to avoid sparks. (**c**) Typical voltage and current waveforms measured by the probes as shown in (**a**). Similar current waveforms are obtained in the two- and three-electrode configurations, except for the peak value. (**d**) Typical current waveforms passing through the resistor, as presented in (**b**). The peak value increases with an increase in DC voltage, indicating that more positive ions are collected and absorbed by the third electrode.*

In order to discuss the discharge structure, a photograph of the discharge in the seven-electrode configuration was taken using the conventional configuration with 2000-Hz pulses applied and an optical gate of 0.1 sec [Figure 8.4(a)]. The discharge spreads from both sides of each exposed electrode because there is a large potential difference between the exposed electrode and the covered electrode at both sides, as discussed above. This discharge structure generates counter ionic-wind, which decelerates the main flow. Contrary to the conventional approach, the discharge occurs and spreads from the downstream side of each exposed electrode with the proposed approach, as shown in [Figure 8.3(b)]. Note that an exposed electrode was divided into two electrodes in this experiment. One electrode was connected to the power supply or ground directly, and the other was connected via a resistor to limit the current. The upstream side of the exposed electrode plays a role in charge absorption and requires a resistor to prevent sparks. Conversely, the downstream side of the exposed electrode plays a role in the ionization of air and cannot ignite the discharge when a high resistive material is connected to the electrode due to the current limitation.

The gap between each divided electrode is 2 mm. A discharge does not occur in this gap because the potential difference between the divided electrodes is small. The distance of 2 mm between each actuator module is quite small compared to conventional multi-electrode DBD plasma actuators. The peak voltage of the third electrode is 300 V, as shown in Figure 8.3(d) (0.3 mA × 1.0 MΩ). The breakdown threshold of air at atmospheric pressure is around 30 kV/cm. [86] Therefore, this distance can be shortened to the order of 0.1 mm in our system. It is interesting to note that the discharge structure above the high-voltage covered electrode is filamentary, whereas the structure above the grounded covered electrode is relatively diffusive. This is caused by the different type of the discharges; cathode- and anode-directed discharges occur above the high-voltage and grounded covered electrodes, respectively.

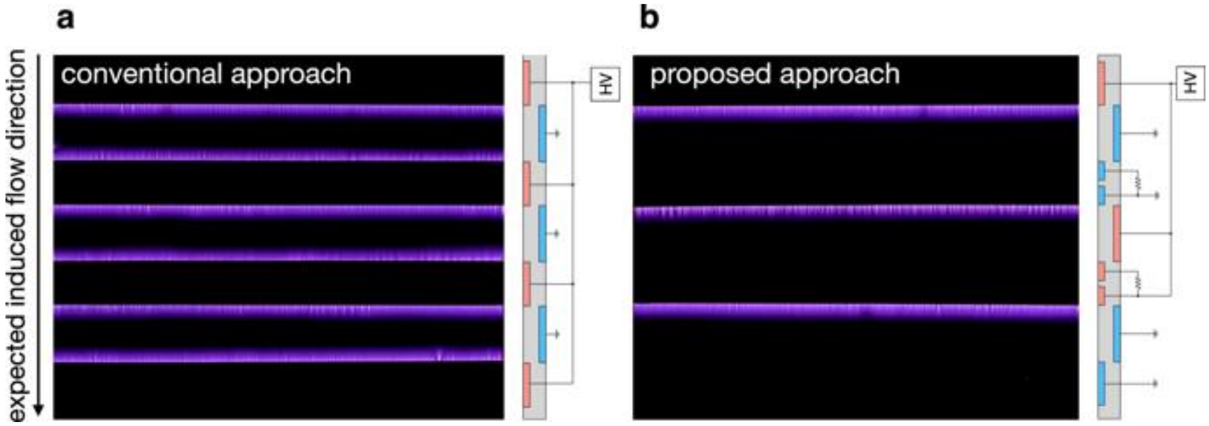

Fig 8.4. *Discharge photographs for (**a**) conventional and (**b**) proposed electrode arrangements using seven electrodes when applying 8-kV DC voltage combining 2000-Hz negative 8-kV pulses. The electrode arrangement is schematically shown on the right-hand side of each photograph. 200 pulses were applied during the camera's exposure. Discharge occurs at both sides of the exposed electrode for (**a**). By contrast, discharge spreads from the upstream side of the exposed electrode for (**b**), preventing unexpected discharge, which generates counter ionic-wind.*

8.3. THRUST MEASUREMENT AND SUCCESSIVE ACCELERATION OF IONIC WIND

The thrust generated by the surface discharge and the induced flow velocity was measured in quiescent air to characterize the performance of the proposed multi-electrode DBD plasma actuator. The effect of the number of electrodes on the thrust parallel to the dielectric surface was measured when 8-kV DC voltage was applied combined with negative 8-kV pulses at a repetitive pulses' frequency of 500, 1000, 2000 Hz [Figure 8.5(a)] For the sake of comparison, the method for counting electrodes is that the two separated electrodes are counted as one electrode.

Although no thrust was measured in the two-electrode case in this study, the thrust increased almost linearly with an increased number of electrodes at all frequencies, consistent with the results of the numerical simulation. This indicates that the degradation of the thrust-generation performance due to the cross-talk phenomenon is suppressed in the proposed approach. The increment of the thrust from the two-electrode to three-electrode cases is caused by the enhancement of the electric field downstream due to the absorption of the surface charge by the exposed electrode. Therefore, this result clearly shows the mutual enhancement (rather than degradation) of the thrust generation between successive actuator modules.

In addition, the thrust increases with an increase to the repetitive pulse frequency, because the generation rate of charged particles increases. The positive EHD force is mainly generated during the dielectric charging phase.[78] The total EHD force corresponds to the sum of the EHD force produced by each nanosecond pulse superposition when the period of the repetitive pulses is longer than the time scale of the positive or negative charging phase. Thus, EHD force generation could start to saturate when the nanosecond pulse is superposed before the dielectric surface is fully charged. However, this time scale is of the order of 0.1 μs, as shown in [Figure 8.2(b)], which corresponds to 10 MHz and is much shorter than the period of repetitive pulses. EHD force generation could also start to saturate when the period of the repetitive pulses becomes shorter than the time required for charge absorption at the upstream side of the exposed electrode. The time constant of charge absorption is roughly 20 μs (i.e., 50 kHz), as shown in [Figure 8.4(b)]. This time constant is also much shorter than the pulse repetition period under our experimental conditions and can be easily reduced by using a low-resistance load. Therefore, a large thrust can be obtained by increasing the repetitive pulse frequency.

The proposed method produces the same order of thrust as a typical AC DBD plasma actuator under similar experimental conditions.[87] The result of the AC DBD plasma actuator is obtained with the same peak-to-peak voltage but with a frequency of 10 kHz. This study mainly aims to provide a highly integrated multi-electrode plasma actuator without generating counter ionic wind. Although the optimization of the electrode gap and the shape of the nanosecond pulse for enhancement of the EHD force generated by the single-module actuator should be addressed, the study of EHD force enhancement in the single-module actuator is beyond the scope of this study and will be performed as future work.

The advantage of our approach over the conventional multi-electrode AC DBD plasma actuator is that each module actuator can be densely arranged, resulting in efficient additional acceleration of the ionic wind. In light of the thrust to power ratio, the efficiency of our approach is the same as or higher than that of a typical AC DBD plasma actuator when we use three or more electrodes, as shown in [Figure 8.5(b)]. The power deposited to air is calculated as described in the previous experimental study [88], which uses a nanosecond pulse voltage. The efficiency of our proposed method approaches around 0.06 mN/W with an increase in the number of electrodes because both the thrust and the power increase when the number of actuator modules increases.

Our single module resembles a conventional three-electrode plasma actuator [89]. Although a performance comparison should be conducted, it will be made as a future work because the present study focuses on the multistage plasma actuator and not on the single-module performance. The main difference between other three-electrode actuators and our single-module actuator is that the DC voltage is applied to the downstream exposed electrode in the conventional method whereas it is superposed on the nanosecond pulses and is applied to the upstream exposed electrode in our method. The idea of using a DC voltage resembles that of the conventional three-electrode configuration because of the acceleration of charged particles. However, the previous three-electrode plasma actuator was used for a single module and cannot be arranged densely. By contrast, the distance between each module can be made quite small in our system.

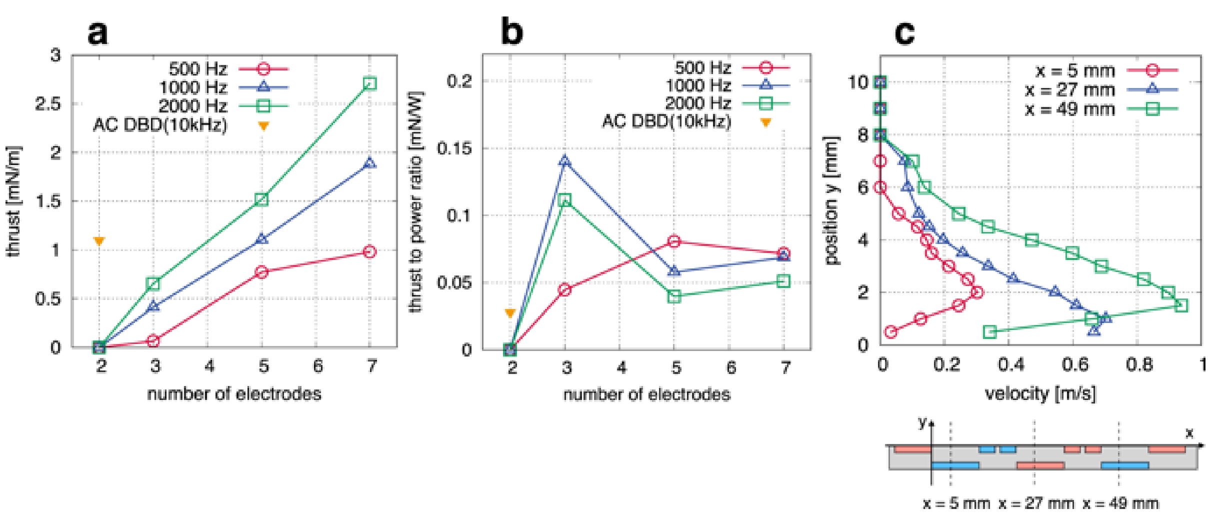

Fig 8.5. *(**a**) Thrust generated by the proposed DBD plasma actuator as a function of the number of electrodes when applying 8-kV DC voltage combining negative 8-kV pulses with a frequency of 500, 1000, and 2000 Hz. This is consistent with the numerical simulation results, and more thrust was obtained at a higher pulse repetition. The thrust produced by a typical AC DBD plasma actuator under similar experimental conditions (except for frequency) [87] is also plotted for comparison. (**b**) Thrust to power ratio as a function of the number of electrodes. The efficiency of our method is higher than that of a typical AC DBD plasma actuator when we use three or more electrodes. (**c**) Stream-wise velocity profiles for the proposed method with a repetitive pulse frequency of 2000 Hz. No apparent flow was obtained for with the conventional method, whereas a wall jet formed and accelerated using the proposed approach.*

8.4. EXISTING SYSTEM AND ITS DISADVANTAGES

Active flow control has the potential to drastically improve the hydrodynamic performance such as suppression of the separating flow and drag reduction, contributing to a reduction of the CO_2 emissions and reduced noise in airplanes. Most active flow-control devices currently available, control the flow by changing the shape of the device using some mechanical moving parts (e.g., leading-edge slats or trailing edge flaps to control the flow around an airfoil). These mechanical flow-control devices can modify the flow and enhance the performance of fluid machinery; however, there are some inherent disadvantages such as complicated components, non-negligible weight increase, low responsiveness.

8.5. PROPOSED SYSTEM AND ITS ADVANTAGES

Active flow control techniques using the electrohydrodynamic (EHD) effect are one of the most promising candidates for next-generation flow-control devices, owing to their advantages over existing active flow control devices – i.e. because they have no moving parts and offer high-speed responsiveness. The EHD effect induces an ionic wind which modifies the flow field around the fluid machinery and generates propulsive force.

Active flow-control devices that use EHD force are referred to as plasma actuators, and they have been studied extensively in the last twenty years. In particular, plasma actuators using dielectric barrier discharge (DBD) are preferred over actuators using DC corona discharge. This is because the transition to an arc discharge, which leads to degradation of the actuator due to heat and renders the power supply unstable, is prevented by the dielectric barrier in the DBD plasma actuators. A single DBD plasma actuator typically consists of two electrodes and one dielectric and generates non-thermal plasma around the exposed electrode (Fig. 1a).

Several researchers have demonstrated that the DBD plasma actuators suppress flow separation and control the boundary layer transition. However, flow control using a DBD plasma actuator at high-speed airflow is confronted by the difficulty of producing sufficient ionic wind for the flow control.

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APPENDIX

The C programming code which was compiled as the User-Defined File (UDF) source in ANSYS Workbench using Visual Studio for the analysis of 2D airfoil as well as the circular cylinder using the DBD Plasma Actuators is given below. This user-defined file contains the necessary user-defined parameters to incorporate the Plasma Actuator onto the 2D airfoil and circular cylinder. This UDF consists of parameters such x and y coordinates of the plasma actuators and the value of the plasma actuator source.

```
//UDF is written in C and compiled by FLUENT using visual studio
//Reference 1: UDF 6.3 User Manual
//Reference 2: West, Thomas, and Serhat Hosder. "Numerical investigation of plasma actuator configurations for flow separation control at multiple angles of attack." 6th AIAA Flow Control Conference. 2012.

//INFORMATION
//define plasma as a rectangle source in momentum to simplify the code

//DECLARATION
//NO validation is done!!

//-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------
#include "udf.h" //udf library in C language
DEFINE_SOURCE(plasma_source, c, t, dS, eqn) //define macro
{
	real xc[ND_ND]; //declaration of cell coordinate
	real source, x, y; //declaration of plasma, x-coordinate and y-coordinate
```

```
    C_CENTROID(xc, c, t);  //centroid of cell
    x = xc[0]; //x as the 1st array of xc
    y = xc[1]; //y as the 2nd array of xc

    if ((x > 0.034) && (y > 0.035) && (x < 0.073) && (y < 0.053)) //define the plasma region
    {
        source = 55000.0; //the value of plasma source
        dS[eqn] = 0; //explicit solution
    }
    else
    {
        source = 0;
        dS[eqn] = 0;
    }
    C_UDMI(c, t, 0) = source; //adding effect
    return source;
}
```

*****

CONCLUSION

Our results demonstrated that the multi-electrode DBD plasma actuator proposed in this study generated mutually enhanced EHD force and successively accelerated the ionic wind, whereas no apparent flow was induced with the conventional approach due to the cross-talk phenomenon, as reported in previous studies. This means that when a landing gear is coupled with such a DBD Plasma Actuator there is significant improvement in the flow around the landing gear which in turn causes less drag and hence reduces noise. The applied voltage waveform is DC voltage combined with repetitive nanosecond pulses forming a two-stroke cycle of the surface charge. The DC component plays a crucial role in accelerating the charged particle, while the nanosecond pulses generate charged particles.

The proposed electrode arrangement prevents the generation of counter ionic-wind because there is no potential difference between the covered electrode and the downstream side of the exposed electrode. Moreover, the upstream side of the exposed electrode absorbs the surface charge and reinforces the electric field, enhancing EHD force instead of degrading the other actuator modules.

Our strategy for strong ionic wind generation differs completely from that of the conventional approach, which requires very high-voltage, making it unsuitable for industrial applications. The results presented in this paper suggest that the applied voltage can be lowered by making the dielectric thinner to the limit of the breakdown voltage. In air, Paschen's law reveals a minimum breakdown voltage of approximately 350 V in a uniform electric field. The EHD force can be increased by increasing the number of electrodes instead of increasing the amplitude of the applied voltage. The actuator module is miniaturized using microelectronic fabrication technology.

The reduction of the driving voltage and further integration of actuator modules will be performed in future work. Moreover, the thrust can be controlled by using DC-bias voltage and pulse repetition frequency independently by using DC-biased repetitive pulses to better control thrust generation.

Our method of integrating DBD plasma actuator modules does not require a high-voltage power supply, which is typically expensive and heavy. Our results suggest that the highly integrated multi-electrode DBD plasma actuator generates strong ionic wind without increasing the voltage amplitude and that it's thus able to drastically improve the aerodynamic performance of aircrafts. The highly integrated DBD plasma actuator is also applicable as an active flow control device using EHD force for various types of hydraulic machinery, because of its low-voltage operation, it easy to use the active flow control device and reduces the cost of introducing a DBD plasma actuator system (in particular, its power supply).

*****

PROPOSED FUTURE WORK

The following is proposed as a research plan for future studies:

(1) Develop DBD plasma actuators with sufficient actuator authority to operate at airspeeds typical of commercial transport landing approach. This will involve the utilization of new dielectric materials like Quartz and Macor.

(2) Obtain more accurate acoustic data for cylinder model and investigate plasma-based noise control for more complicated landing gear models mentioned earlier. To conduct new microphone measurements in an anechoic open jet wind tunnel.

(3) Develop performance and efficiency metrics for DBD plasma flow control. Using existing data compare the performance and efficiency of DBD plasma flow control with passive vortex generators installed on the cylinder surface. Quantify advantages and limitations, if any, on the application of DBD flow control and airframe noise reduction for flight speeds of commercial transport aircraft in landing approach or take-off configurations.

(4) Perform experimental validations of the Large Eddy Simulation (LES) based computational tool for the design of plasma-based airframe noise control strategies. LES modelling of plasma flow control experiments for single and tandem cylinders. Quantify uncertainties in experiments and predictions and differences between experimental results and predictions.

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